

FUNCTIONAL REFERENCE MANUAL

# F&B Backend Structure Module

Master Configuration, Recipes, Kitchen, Service, Inventory and Analytics

*Six-board functional architecture with standard POS coverage*

VERSION 1.0

August 2026

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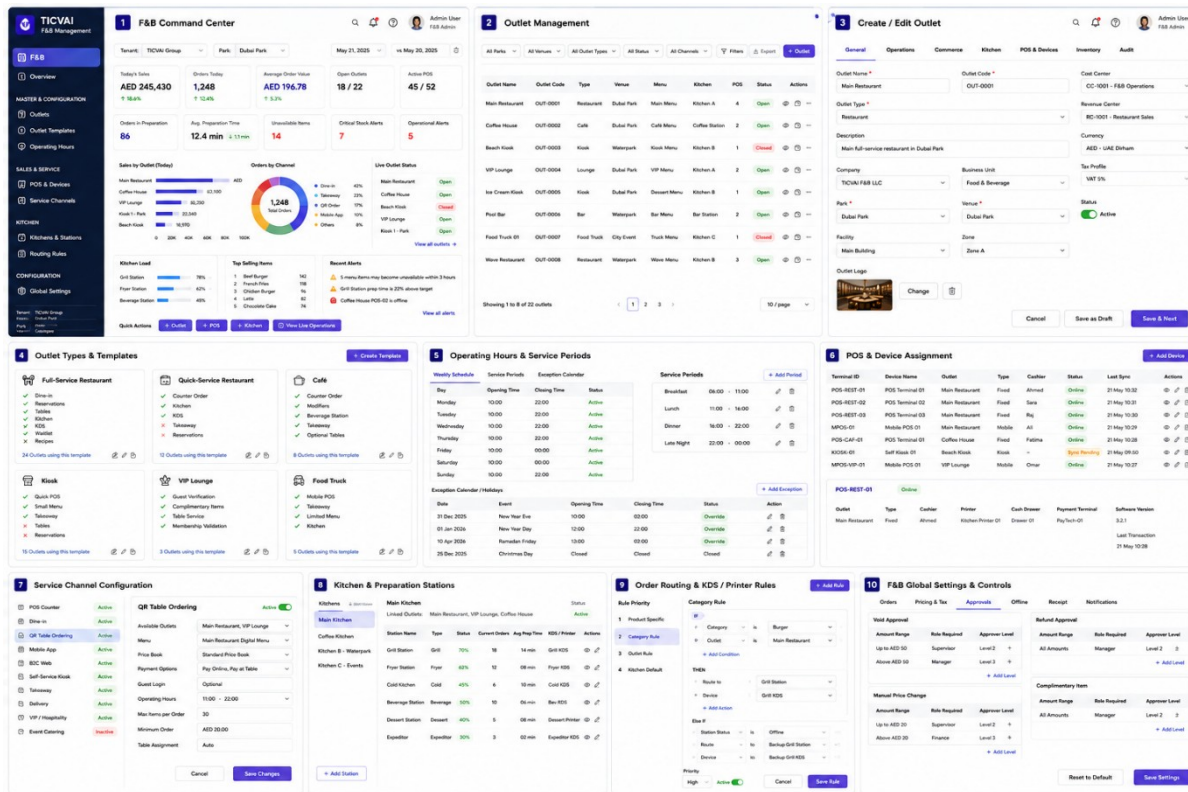
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# F&B — Board 1: Master & Outlet Configuration



**Purpose:** Give the F&B administrator one place to configure the operational structure before menus, recipes, kitchen operations, tables, or inventory are configured.

## Screen 1 — F&B Command Center

### Top KPI cards

- Today's Sales
- Orders Today
- Average Order Value
- Open Outlets
- Active POS
- Orders in Preparation
- Average Preparation Time
- Unavailable Items
- Critical Stock Alerts
- Operational Alerts

### Main widgets

- Sales by Outlet
- Orders by Channel
- Live Outlet Status
- Kitchen Load
- Top Selling Items
- Recent Alerts

AI Insight panel

Main Restaurant is 22% busier than a typical Wednesday.  
Grill Station preparation time has increased from 8 to 14 minutes.  
5 menu items may become unavailable within 3 hours.

Quick actions

+ Outlet | + POS | + Kitchen | View Live Operations  
This screen should be operational rather than purely financial.

Screen 2 — Outlet Management

This is the master list of all F&B outlets.

Filters

Tenant | Company | Park | Venue | Outlet Type | Status | Service Channel

Table

| Outlet          | Type       | Venue     | Menu       | Kitchen      | POS | Status |
|-----------------|------------|-----------|------------|--------------|-----|--------|
| Main Restaurant | Restaurant | Park A    | Main Menu  | Kitchen A    | 4   | Open   |
| Coffee House    | Café       | Park A    | Café Menu  | Café Station | 2   | Open   |
| Beach Kiosk     | Kiosk      | Waterpark | Kiosk Menu | Kitchen B    | 1   | Closed |
| VIP Lounge      | Lounge     | Park A    | VIP Menu   | Kitchen A    | 2   | Open   |

Actions:

View | Edit | Duplicate | Suspend | Archive

Important feature

I would include **Duplicate Outlet**.  
If a customer operates 20 almost-identical kiosks, they shouldn't configure all 20 manually.

Screen 3 — Create / Edit Outlet

This is the detailed configuration screen.  
I would use tabs.

Tab 1 — General

- Outlet Name
- Outlet Code
- Outlet Type
- Description
- Company
- Business Unit
- Park
- Venue
- Facility
- Zone
- Cost Center
- Revenue Center

- Currency
- Status

**Tab 2 — Operations**

- Opening Hours
- Service Periods
- Capacity
- Dine-In
- Takeaway
- Mobile Ordering
- QR Ordering
- Delivery
- Reservations

**Tab 3 — Commerce**

- Default Menu
- Price Book
- Tax Profile
- Service Charge
- Accepted Payment Methods
- Promotion Rules
- Loyalty Rules

**Tab 4 — Kitchen**

- Primary Kitchen
- Preparation Stations
- KDS
- Kitchen Printers
- Routing Profile

**Tab 5 — POS**

Assigned terminals and devices.

**Tab 6 — Inventory**

- Inventory Location
- Stock Location
- Replenishment Source
- Default Warehouse

Notice that this only **links** the outlet to Shared Inventory. It does not create another inventory engine.

**Tab 7 — Audit**

Created by

Modified by

Approved by

Version history

The matrix requires configuration changes and sensitive actions to be auditable, so this should be standard throughout the module.

## Screen 4 — Outlet Types & Templates

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This is one of the screens I think will make TICVAI considerably easier to configure.

### Templates

#### Full-Service Restaurant

- ✓ Dine-In
- ✓ Reservations
- ✓ Tables
- ✓ Kitchen
- ✓ KDS
- ✓ Waitlist
- ✓ Recipes

#### Quick-Service Restaurant

- ✓ Counter Order
- ✓ Kitchen
- ✓ KDS
- ✓ Takeaway
- × Reservations

#### Café

- ✓ Counter Order
- ✓ Modifiers
- ✓ Beverage Station
- ✓ Takeaway
- ✓ Optional Tables

#### Kiosk

- ✓ Quick POS
- ✓ Small Menu
- ✓ Takeaway
- × Tables
- × Reservations

#### VIP Lounge

- ✓ Guest Verification
- ✓ Complimentary Items
- ✓ Table Service
- ✓ Membership/Entitlement Validation

### Buttons

+ Create Template

Duplicate

Edit

Assign to Outlets

This is especially useful for TICVAI's multi-venue/multi-tenant model.

## Screen 5 — Operating Hours & Service Periods

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This screen manages **when the outlet and menus operate**.

### Weekly calendar

| Day     | Opening | Closing | Service Periods             |
|---------|---------|---------|-----------------------------|
| Monday  | 10:00   | 22:00   | Lunch / Dinner              |
| Tuesday | 10:00   | 22:00   | Lunch / Dinner              |
| Friday  | 10:00   | 00:00   | Lunch / Dinner / Late Night |

Service periods

Breakfast

08:00–11:00

Lunch

11:00–16:00

Dinner

16:00–22:00

Late Night

22:00–00:00

But I would add something important:

Exception Calendar

Ramadan

Eid

Christmas

Special Event

Private Event

Maintenance

Venue Closure

Example:

31 December

Normal: 10:00–22:00

Override: **10:00–02:00**

Menus, prices and availability can later reference these periods.

Screen 6 — POS & Device Assignment

This screen connects physical POS infrastructure to F&B.

Device cards/table

| Terminal | Outlet          | Type   | Cashier | Payment | Printer    | Status |
|----------|-----------------|--------|---------|---------|------------|--------|
| POS-01   | Main Restaurant | Fixed  | Ahmed   | Online  | Printer 01 | Online |
| POS-02   | Main Restaurant | Fixed  | Sara    | Online  | Printer 02 | Online |
| MPOS-04  | VIP Lounge      | Mobile | Omar    | Online  | —          | Online |

Terminal configuration

- Terminal ID
- Device Name
- Outlet
- Default Menu

- Price Book
- Cash Drawer
- Payment Terminal
- Receipt Printer
- Kitchen Printer
- Customer Display
- Default Language
- Offline Mode
- Cashier Assignment
- Concurrent Login Rules

The original matrix specifically requires POS/back-office login controls, supervisor override and configurable concurrent terminal access.

### **Device health**

I would also display:

Online  
Sync Pending  
Offline

plus:

Last Sync  
Software Version  
Last Transaction

## **Screen 7 — Service Channel Configuration**

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This controls **how customers can order**.

### **Channels**

#### **POS Counter**

#### **Dine-In**

#### **QR Table Order**

#### **Mobile App**

#### **B2C Web**

#### **Self-Service Kiosk**

#### **Takeaway**

#### **Delivery**

#### **VIP / Hospitality**

#### **Event Catering**

Clicking a channel opens its rules.

### **Example — QR Table Ordering**

Status: Active

Available Outlets:

Main Restaurant

Menu:

Main Restaurant Digital Menu

Price Book:

Standard

Payment:

Pay Online / Pay at Table

Guest login:

Optional

Operating hours:

11:00–22:00

Maximum items/order:

30

Minimum order:

AED 20

This architecture also allows a product to be available in one channel but not another, consistent with the matrix's channel/product assignment concept.

## Screen 8 — Kitchen & Preparation Stations

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Now we configure the physical production environment.

### Kitchen hierarchy

#### Main Kitchen

→ Grill Station

→ Fry Station

→ Cold Kitchen

→ Dessert Station

→ Beverage Station

→ Expeditor

Each station shows:

Status

Current Orders

Average Preparation Time

KDS Device

Printer

Staff Assigned

#### Create Station

Fields:

- Station Name
- Station Code
- Kitchen
- Outlet(s)
- Product Categories
- Preparation Capacity
- Target Preparation Time
- KDS
- Backup Station
- Printer
- Operating Hours

### Important feature — shared kitchen

One kitchen must be able to support **multiple outlets**.

Example:

Main Kitchen



Restaurant   VIP Lounge   Kiosk

## Screen 9 — Order Routing & KDS/Printer Rules

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Screen 8 creates the stations.

Screen 9 determines **where orders go**.

### Visual Routing Builder

For example:

#### Category = Burger

→ Grill Station

→ Grill KDS

#### Category = Fries

→ Fry Station

→ Fry KDS

#### Category = Coffee

→ Beverage Station

→ Beverage KDS

### More advanced rules

IF:

Outlet = VIP Lounge

AND Product Category = Dessert

THEN:

→ VIP Kitchen Station

Or:

IF Grill Station = Offline

THEN:

→ Backup Grill Station

This should be a **no-code rule builder**.

### Routing priority

1. Product-specific rule
2. Category rule
3. Outlet rule
4. Kitchen default

That prevents configuration conflicts.

## Screen 10 — F&B Global Settings & Controls

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The final screen controls F&B-wide behavior.

I would divide it into tabs.

### **Orders**

- Allow Hold Order
- Allow Reopen
- Allow Split Bill
- Allow Merge Bill
- Allow Item Transfer
- Allow Table Transfer
- Allow Void
- Allow Refund
- Maximum Open Order Duration

### **Pricing & Tax**

- Tax Inclusive / Exclusive
- Default VAT
- Service Charge
- Municipality Fee
- Rounding
- Gratuity

### **Approvals**

This is particularly important for TICVAI.

Configure approval thresholds:

#### **Void**

Under AED 50 → Supervisor

Above AED 50 → Manager

#### **Refund**

Manager Approval

#### **Manual Price Change**

Finance Approval

#### **Complimentary Item**

Manager Approval

#### **Discount Override**

Supervisor Approval

The matrix explicitly requires configurable approval requirements and separation of duties around refunds, promotions, price changes and manual overrides.

### **Offline**

- Allow Offline Sales
- Maximum Offline Transaction Value
- Offline Payment Rules
- Synchronization Rules

### **Receipt**

- Logo
- Header
- Footer
- VAT Number

- QR
- Language
- Digital Receipt

### How I would visually organize Board 1

The left navigation could look like:

FOOD & BEVERAGE

▣ Overview

MASTER & CONFIGURATION

Outlets

Outlet Templates

Operating Hours

SALES & SERVICE

POS & Devices

Service Channels

KITCHEN

Kitchens & Stations

Routing Rules

CONFIGURATION

Global Settings

And the header should always contain:

#### **Tenant → Company → Park/Venue → Outlet**

This hierarchy is critical because TICVAI is multi-tenant and multi-venue.

### One AI feature I would add to Board 1

I don't want AI only appearing in Board 6.

Put an **AI Configuration Assistant** directly into Board 1.

For example, the administrator writes:

Create a quick-service burger restaurant in Park A. Opening hours are 10 AM–11 PM. It has four POS terminals, one grill station, one fryer station and one beverage station. It supports dine-in, takeaway and QR ordering.

TICVAI generates a proposed configuration:

**Outlet:** Burger House

**Template:** Quick Service

**Hours:** 10:00–23:00

**POS:** 4

**Kitchen Stations:** 3

**Channels:** Dine-In / Takeaway / QR

**Suggested routing:** Generated

Then:

#### **Review Configuration → Submit for Approval → Publish**

That aligns much more strongly with TICVAI's AI-first positioning than simply adding an AI chatbot later.

### Board 1 workflow

The complete configuration workflow becomes:

#### **Create Outlet**

→ Choose Template

- Configure Operating Hours
- Assign Menu/Price Book
- Assign POS
- Enable Service Channels
- Configure Kitchen
- Configure Routing
- Configure Controls
- Validate Configuration
- Approval
- **Publish**

I would also use the lifecycle:

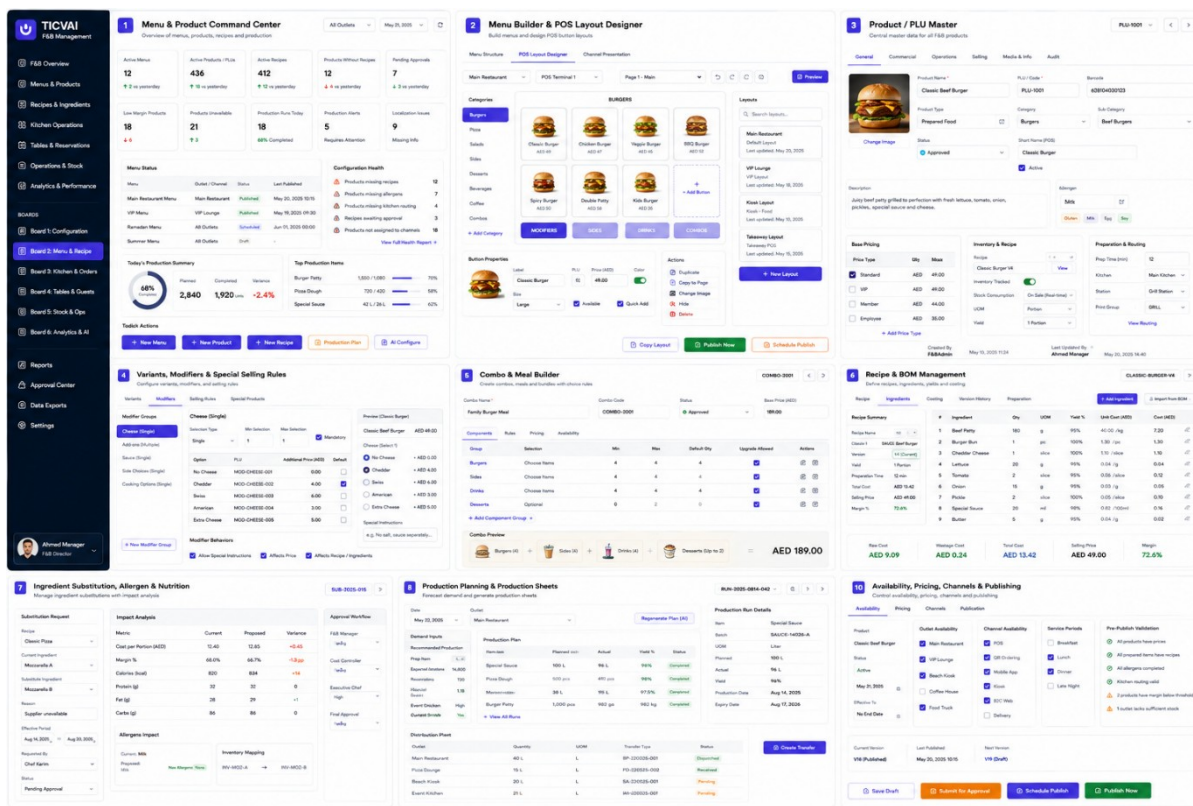
**Draft → Review → Approved → Published → Suspended → Archived**

so administrators aren't making live F&B changes accidentally.

With this structure, **Board 1 establishes the entire F&B operating environment**, while Board 2 can focus completely on **Menu, Product, Modifier & Recipe Management** without duplicating outlet configuration.

# REVISED F&B — BOARD 2

## Menu, Product, Recipe & Production Configuration



Board 2 should manage four connected layers:

- What we sell** → Menu & Product
- How guests customize it** → Variants & Modifiers
- How we make it** → Recipe & Ingredients
- What we need to prepare** → Production & Commissary

## Revised 10 Screens

| # | Screen                                      | Main Function                                       |
|---|---------------------------------------------|-----------------------------------------------------|
| 1 | Menu & Product Command Center               | Overall menu/product/recipe/production status       |
| 2 | Menu Builder & POS Layout Designer          | Menus + POS button layouts + channel presentation   |
| 3 | Product / PLU Master                        | Central F&B product configuration                   |
| 4 | Variants, Modifiers & Special Selling Rules | Sizes, choices, modifiers, limits, special products |
| 5 | Combo & Meal Builder                        | Meals, bundles and choice-based combinations        |

| #  | Screen                                        | Main Function                                   |
|----|-----------------------------------------------|-------------------------------------------------|
| 6  | Recipe & BOM Management                       | Recipes, ingredients, yields, costing, versions |
| 7  | Ingredient Substitution, Allergen & Nutrition | Controlled substitutions and food information   |
| 8  | Production Planning & Production Sheets       | Demand-based kitchen preparation planning       |
| 9  | Central Kitchen & Commissary Management       | Central production supplying multiple outlets   |
| 10 | Availability, Pricing, Channels & Publishing  | Where/when/how products are sold and published  |

## SCREEN 1 — Menu & Product Command Center

It should become the **control center for Board 2**.

### KPI cards

- Active Menus
- Active Products / PLUs
- Active Recipes
- Products Without Recipes
- Products Without Kitchen Routing
- Pending Recipe Approvals
- Low-Margin Products
- Products Unavailable
- Production Runs Today
- Production Alerts

### Menu status

Show:

Main Restaurant Menu — Published

VIP Menu — Published

Ramadan Menu — Scheduled

Summer Menu — Draft

### Configuration health

TICVAI should automatically identify:

**12 products missing recipes**

**7 products missing allergens**

**4 products missing kitchen routing**

**3 recipes awaiting approval**

**18 products not assigned to digital channels**

## Production summary

Today's Planned Production:

1,450 Burger Patties  
680 Pizza Dough  
42 L Special Sauce

Production completion:

**68%**

## Quick actions

**+ Menu | + Product | + Recipe | + Production Plan | AI Configure**

## SCREEN 2 — Menu Builder & POS Layout Designer

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This screen contain **two related but separate modes**.

### Mode A — Menu Structure

Example:

#### Main Restaurant Menu

Burgers  
→ Classic Burger  
→ Chicken Burger  
→ Veggie Burger

Pizza  
→ Margherita  
→ Pepperoni

Beverages  
→ Soft Drinks  
→ Coffee  
→ Juice

Administrators can drag and drop products and categories.

### Mode B — POS Layout Designer

The additional matrix explicitly requires authorized users to remotely control product visibility and placement on POS screens, apply changes immediately or at a future time, and reuse menu layouts across locations.

So we need a genuine visual POS designer:

```
┌────────── POS LAYOUT ─────────┐  
[BURGERS] [PIZZA] [SALADS] [DRINKS]  
[Classic] [Chicken] [Veggie] [Combo]  
[Fries] [Water] [Coke] [Coffee]  
+ ADD BUTTON
```

Admin can:

- Drag buttons
- Resize buttons
- Change product position
- Create category pages
- Hide products
- Highlight featured products
- Copy layouts

### Copy layout

#### Main Restaurant POS

→ Copy to:

- ☒ Restaurant 2
- ☒ Restaurant 3
- ☒ VIP Lounge

#### Publication

#### Publish Now

or

**Schedule: 01 Sep 2026 — 06:00**

This needs version control and rollback.

## SCREEN 3 — Product / PLU Master

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### Product information

- Product Name
- Product Code
- PLU
- Barcode
- Product Type
- Category
- Subcategory
- Brand
- Description
- POS Short Name
- Image
- Status

The earlier matrix also specifies predefined naming schemas/code formats, which should be configurable rather than manually enforced.

Example:

**DXB-FNB-BURGER-001**

## Commercial

- Base Price
- Cost
- Currency
- VAT
- Service Charge
- Revenue Center
- Cost Center
- Accounting Mapping

## Operations

- Preparation Required
- Recipe
- Preparation Time
- Kitchen
- Station
- Inventory Tracked
- Stock Consumption Method

## Selling

- Outlet
- Menu
- Channel
- Service Period
- Customer Segment
- Membership
- Group Type

## Product status

Draft  
Review  
Approved  
Published  
Temporarily Unavailable  
Suspended  
Archived

---

## SCREEN 4 — Variants, Modifiers & Special Selling Rules

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### Variants

Example:

#### Pizza

Small — AED 35  
Medium — AED 45  
Large — AED 58  
Family — AED 72

Each can have separate:

PLU  
Price  
Recipe  
Cost  
Preparation Time  
Availability.

## **Modifier groups**

### **Cheese**

No Cheese  
Cheddar + AED 4  
Swiss + AED 6

### **Sides**

Fries  
Salad  
Sweet Potato + AED 5

### **Modifier controls**

Mandatory / Optional  
Minimum Selection  
Maximum Selection  
Default  
Additional Price  
Ingredient Consumption  
Kitchen Instruction.

The additional matrix explicitly requires mandatory/optional modifiers and free-text special requests where modifiers are insufficient.

So add:

### **Special Request**

"No salt, sauce separately."

### **Selling limits**

This is also where I would configure:

#### **Maximum per Order**

#### **Maximum per Day**

#### **Maximum per Time Slot**

#### **Maximum per Customer**

#### **Group-Only Product**

#### **School Product**

## Event Product

### Happy Hour

Example:

### School Meal

Customer Segment:  
School Group

Available:  
10:00–14:00

Maximum:  
500/day

## SCREEN 5 — Combo & Meal Builder

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### Example

#### Family Burger Meal — AED 189

Choose:

#### 4 Burgers

#### 4 Sides

#### 4 Drinks

#### Component groups

Main  
Side  
Drink  
Dessert  
Add-On

#### Rules

Required quantity  
Minimum  
Maximum  
Substitution allowed  
Upgrade allowed  
Price difference  
Modifier eligibility.

#### Pricing

Fixed Price  
Component Price  
Bundle Discount

Upgrade Price  
Promotional Price.

**Inventory**

The actual components selected must drive inventory consumption.

So if the customer chooses:

3 Beef Burgers  
1 Chicken Burger

TICVAI consumes the corresponding recipes—not a generic "Family Meal" inventory item.

**SCREEN 6 — Recipe & BOM Management**

The new matrix explicitly requires recipe creation/change tracking, approval before changes become active, production planning, batch preparation and production runs.

**Recipe header**

**Classic Beef Burger**

Version:  
V4

Yield:  
1 Portion

Preparation:  
12 min

Cost:  
AED 13.42

Selling:  
AED 49

Margin:  
72.6%

**Ingredients**

| Ingredient | Qty | UOM   | Cost | Waste/Yield |
|------------|-----|-------|------|-------------|
| Beef Patty | 180 | g     | 7.20 | 95%         |
| Bun        | 1   | pc    | 1.30 | 100%        |
| Cheese     | 1   | slice | 1.10 | 100%        |
| Sauce      | 20  | ml    | 0.82 | 98%         |

**Recipe versions**

V1  
V2  
V3  
V4 Current

Compare:

**V3 ↔ V4**

## **Workflow**

Draft

→ Testing

→ Cost Review

→ F&B Approval

→ Finance/Cost Approval if required

→ Active

## **Yield & shrinkage**

Raw Beef:  
10 kg

Usable after preparation:  
9.2 kg

Yield:  
**92%**

TICVAI should calculate the real usable cost.

## **Recipe Cost Revaluation**

Authorized users shall be able to revalue every ingredient in a recipe using either the Last Receipt Rate or the Last Issue Rate.

Last Receipt Rate: the latest approved procurement rate at which the ingredient was received from a supplier.

Last Issue Rate: the latest approved rate at which the ingredient was issued from the store or warehouse to the kitchen or production location.

The selected valuation basis may be configured at company, outlet, recipe or ingredient level, subject to role-based permissions and approval controls.

Before applying a revaluation, the system shall show the previous rate, proposed rate, effective date, source receipt or issue transaction, revised recipe cost, revised margin and affected products.

Revaluation shall support preview, approval, effective dating and bulk recalculation, while retaining the previous value and a complete audit trail. It shall not rewrite historical closed transactions unless an authorized financial adjustment process is used.

## Nested Recipes / Sub-Recipes

A recipe shall be allowed as an ingredient or component within another recipe. For example, a Special Sauce recipe may be linked as a sub-recipe within a Burger recipe.

The parent recipe shall reference the approved sub-recipe version, required quantity and unit of measure, and shall automatically roll up the sub-recipe's ingredient cost, yield, wastage, allergens, nutrition and theoretical inventory consumption.

Changes to an active sub-recipe shall identify all affected parent recipes and products, recalculate their costs and margins, and follow the configured approval and effective-date workflow before publication.

The system shall prevent circular recipe links and provide a multi-level recipe-explosion view from the finished product down to the lowest-level ingredients.

## Open Item Recipe and Receipt Linkage

The system shall provide an Open Item Recipe option for POS items that are not selected from a predefined product/PLU master record.

When an open item is sold, the cashier shall be able—or, where configured, be required—to link the sale to an approved open-item recipe or recipe template before settlement.

The open-item receipt line shall retain the entered item description, quantity, unit of measure, selling price, tax, linked recipe, calculated cost and margin, subject to permissions and approval thresholds.

The linked recipe shall drive ingredient-level theoretical consumption, costing, stock deduction, reporting and audit history. If no recipe is linked, the transaction shall be flagged as an exception for authorized review and reconciliation.

## SCREEN 7 — Ingredient Substitution, Allergen & Nutrition

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The matrix requires approved substitutions to automatically update **recipe cost, allergens, nutrition, inventory consumption and audit logs**.

### Example

Recipe:

Classic Pizza

Current Ingredient:

**Mozzarella A**

Proposed substitute:

**Mozzarella B**

### Impact Analysis

Cost:

AED 12.40 → **AED 12.85**

Margin:

68.0% → **66.7%**

Calories:

820 → **834**

Allergens:

Milk ✓

No new allergen

Inventory Mapping:

INV-MOZ-A

→

INV-MOZ-B

**Approval**

Reason:

Supplier unavailable

Effective:

Temporary — 14–20 Aug

Approval:

F&B Manager → Cost Controller

Buttons:

**Approve | Reject | Modify**

**Allergens**

Contains:

Milk

Egg

Wheat

May contain:

Nuts

**Dietary**

Vegetarian

Vegan

Halal

Gluten-Free  
Dairy-Free etc.

Every approved change should propagate to POS, mobile, QR and kiosk product information.

## SCREEN 8 — Production Planning & Production Sheets

The matrix specifically requires production quantities, AI forecasting, production sheets and schedules using reservations, attendance forecasts, events and inventory availability.

### Production forecast

#### Tomorrow — Dubai Park

Expected Attendance:

14,800

Restaurant Reservations:

720

Expected F&B Orders:

5,328

### Recommended production

| Prep Item     | Forecast Demand | Safety | Required | Available | Produce |
|---------------|-----------------|--------|----------|-----------|---------|
| Burger Patty  | 1,480           | 70     | 1,550    | 320       | 1,230   |
| Pizza Dough   | 680             | 40     | 720      | 180       | 540     |
| Special Sauce | 38 L            | 4 L    | 42 L     | 12 L      | 30 L    |

### Inputs

TICVAI should show WHY:

Ticket Forecast  
Reservations  
Historical Demand  
Event Schedule  
Current Stock  
Incoming Stock.

### Production Sheet

Click:

**Generate Production Sheet**

creates:

Kitchen  
Station  
Prep Item  
Quantity  
Start Time  
Required By  
Batch  
Staff/Team  
Status.

**Lifecycle**

Planned → Approved → Released → In Production → Completed → Variance Review

**SCREEN 9 — Central Kitchen & Commissary Management**

---

The additional matrix explicitly calls for central kitchen/commissary operations supplying multiple outlets with production batches, transfers, yields and planning.

**Central Kitchen Dashboard**

Today:

Production Runs:  
18

Planned:  
2,840 units

Completed:  
1,920

Variance:  
-2.4%

Outbound Transfers:  
14

**Production run**

**RUN-2026-0814-042**

Item:

Special Sauce

Planned:  
100 L

Actual:  
96 L

Yield:

96%

Batch:

SAUCE-140826-A

Expiry:

17 Aug 2026

### **Distribution**

Main Restaurant:

40 L

VIP Lounge:

15 L

Beach Kiosk:

20 L

Event Kitchen:

21 L

### **Workflow**

Demand Forecast

→ Production Plan

→ Central Kitchen

→ Batch Production

→ Quality/Release

→ Shared Inventory

→ Outlet Transfer

→ Outlet Receiving

### **Important boundary**

Board 2 manages the **production requirement and commissary operation**.

The actual enterprise stock ledger and warehouse transfer remain within Shared Inventory.

---

## **SCREEN 10 — Availability, Pricing, Channels & Publishing**

---

### **Product**

Classic Burger

### **Outlets**

Main Restaurant ✓  
VIP Lounge ✓  
Beach Kiosk ✓  
Coffee House ✕

## Channels

POS ✓  
QR ✓  
Mobile ✓  
Kiosk ✓  
B2C ✓  
Delivery ✕

## Service periods

Lunch ✓  
Dinner ✓  
Late Night ✕

## Pricing

Standard:  
AED 49

VIP:  
AED 49

Member:  
AED 44

Employee:  
AED 35

Happy Hour:  
AED 42

## Availability rules

Stock-Based ✓

Recipe-Based ✓

Kitchen Capacity ✓

Date/Time ✓

## Publishing

Current Version:

V18

Proposed:

V19

Changes:

14 Products  
3 Prices  
2 Recipes  
1 Layout

Affected:

6 Outlets  
28 POS  
4 Digital Channels

### Pre-Publish Validation

TICVAI checks:

- ✓ All products have prices
- ✓ All prepared items have recipes
- ✓ All allergens completed
- ✓ Kitchen routing valid
- △ 2 products have margin below threshold
- △ 1 outlet lacks sufficient stock

Then:

### Submit for Approval

→ **Schedule**

→ **Publish**

### AI Configuration Assistant

AI now needs to understand the entire Board 2 configuration.

For example:

Create a new Ramadan Iftar menu for Main Restaurant and VIP Lounge from 1 March to 30 March, 5:30 PM–10 PM. Include three meal packages, route grills to Main Kitchen, generate the required recipes and estimate tomorrow's production from our reservations and ticket forecast.

TICVAI can create a **draft proposal**, not publish automatically:

Menu  
Products  
Combos  
Modifiers  
Recipes  
Kitchen routing

Availability  
POS layout  
Production requirements.

Then:

**Review → Validate → Approval → Publish**

## **Board 2 architecture**

This is now much more complete:

### **MENU**



### **PRODUCT / PLU**



### **VARIANT + MODIFIER + COMBO**



### **RECIPE**



### **INGREDIENT + SUBSTITUTION + ALLERGEN**



### **DEMAND**



### **PRODUCTION PLAN**



### **CENTRAL KITCHEN / COMMISSARY**



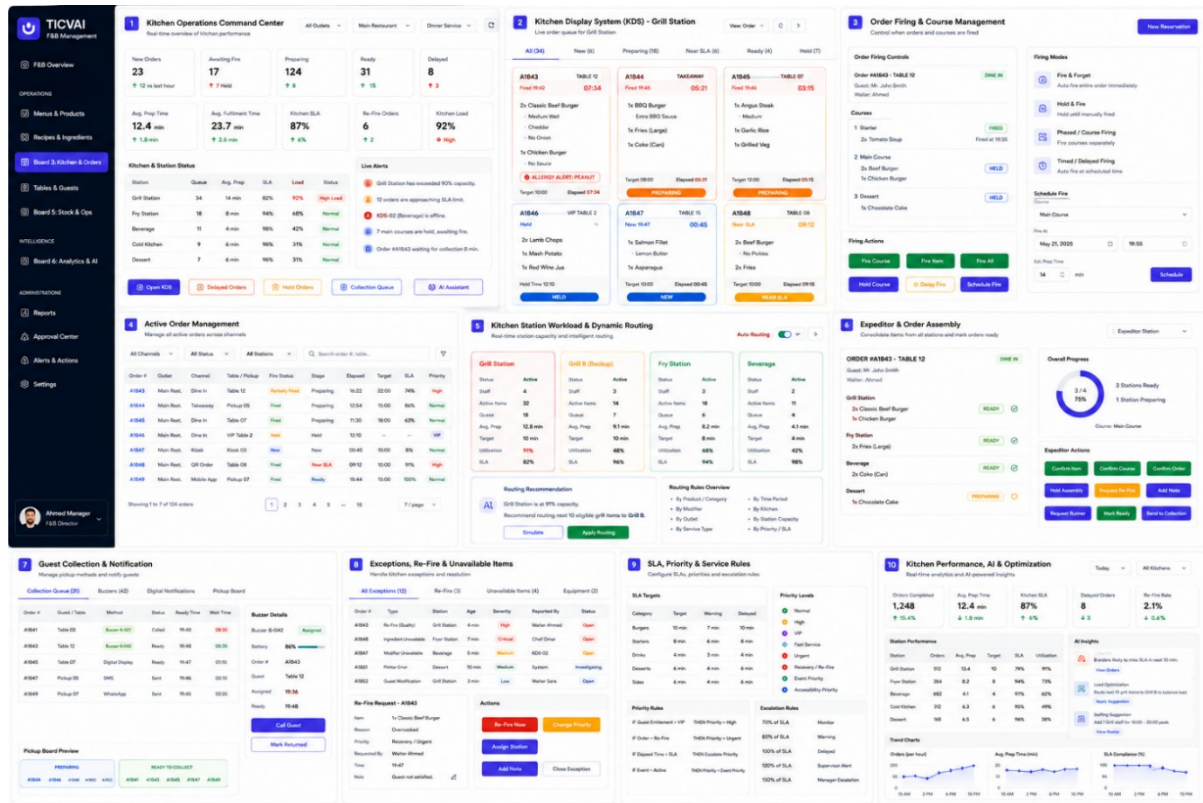
### **AVAILABILITY + PRICE + CHANNEL**



### **POS / QR / MOBILE / KIOSK / B2C**

# TICVAI F&B — BOARD 3

## Kitchen, KDS & Order Fulfilment Management — Scope of Work



### 1. Purpose

The Kitchen, KDS & Order Fulfilment Board shall provide a centralized, real-time operational platform for managing the complete lifecycle of Food & Beverage orders from the moment an order is confirmed until preparation, assembly, guest notification, collection/service, and completion.

The solution should support orders originating from:

- Fixed POS
- Mobile POS
- Self-Service Kiosk
- QR Table Ordering
- Mobile Application
- B2C Web
- Dine-In
- Takeaway
- VIP / Hospitality
- Group / School Orders
- Event Catering
- Pre-Orders
- Future third-party delivery integrations

All channels shall feed into a **single F&B Order & Fulfilment Engine**.

The operational lifecycle shall support:

Order Received → Validation → Firing Decision → Kitchen Routing → Station Queue → Preparation → Expediting → Ready → Guest/Staff Notification → Served/Collected → Completed

The module shall support advanced kitchen operations including **Hold & Fire, Fire Now, phased/course firing, timed firing, dynamic kitchen routing, KDS, kitchen printers, preparation SLAs, priority orders, buzzers, digital collection notifications, exception management and AI-assisted kitchen optimization.**

## SCREEN 1 — Kitchen Operations Command Center

### Purpose

Provide kitchen supervisors, restaurant managers and F&B management with a real-time overview of all kitchen and fulfilment operations.

### Global Filters

- Tenant
- Company
- Park
- Venue
- Outlet
- Kitchen
- Station
- Service Period
- Order Channel

### KPI Cards

- New Orders
- Orders Awaiting Fire
- Orders Preparing
- Orders Ready
- Delayed Orders
- Average Preparation Time
- Average Fulfilment Time
- Kitchen SLA %
- Re-Fire Orders
- Active Stations
- Offline Stations
- Current Kitchen Load

### Kitchen Status

| Kitchen / Station | Queue | Avg. Prep | SLA | Load | Status    |
|-------------------|-------|-----------|-----|------|-----------|
| Grill Station     | 34    | 14 min    | 82% | 92%  | High Load |
| Fry Station       | 18    | 8 min     | 94% | 68%  | Normal    |
| Beverage          | 11    | 4 min     | 98% | 42%  | Normal    |
| Dessert           | 7     | 6 min     | 96% | 31%  | Normal    |

### Live Alerts

Examples:

Grill Station has exceeded 90% capacity.

12 orders are approaching their SLA limit.

Beverage KDS-02 is offline.

7 main courses are currently held awaiting Fire instruction.

Order #A1843 has been waiting for collection for 8 minutes.

### Quick Actions

**Open KDS | Delayed Orders | Held Orders | Activate Backup Station | Collection Queue | AI Operations Assistant**

## SCREEN 2 — Kitchen Display System (KDS)

---

### Purpose

Provide kitchen teams with a fast, touch-optimized operational interface for receiving and preparing orders.

The KDS shall support both touch displays and compatible physical bump-bar controls.

### Order Card

Each KDS order card shall display:

- Order Number
- Outlet
- Order Channel
- Table / Collection Number
- Order Time
- Fire Time
- Elapsed Preparation Time
- Target Preparation Time
- Priority
- Course
- Products
- Quantity
- Variants
- Modifiers
- Special Instructions
- Allergy Alerts
- Guest / Service Notes
- Current Status

Example:

**ORDER #A1843 — TABLE 12**

**MAIN COURSE — FIRED 19:42**

2 × Classic Beef Burger  
• Medium Well

- Cheddar
- No Onion

1 × Chicken Burger

- No Sauce

### **ALLERGY ALERT: PEANUT**

Target: 10 min  
Elapsed: 07:34

### **KDS Statuses**

- New
- Accepted
- Held
- Fired
- Preparing
- Near SLA
- Delayed
- Ready
- Re-Fire
- Cancelled

### **KDS Actions**

Kitchen staff shall be able to:

#### **Accept**

#### **Start Preparation**

#### **Hold**

#### **Mark Ready**

#### **Recall**

#### **Re-Fire**

#### **Escalate**

#### **Add Note**

#### **Report Ingredient Unavailable**

#### **Request Supervisor**

### **KDS Views**

The interface should support:

#### **Order View**

#### **Item View**

**Station View**

**Course View**

**Priority View**

## SCREEN 3 — Order Firing & Course Management

---

### Purpose

Provide advanced control over **when kitchen preparation should begin**.

This is particularly important for full-service restaurants, VIP dining, hospitality and event dining.

The system shall support four primary firing modes:

#### 1. Fire & Forget

The complete order is automatically sent to the kitchen immediately after confirmation.

Example:

Quick-service restaurant order

→ Payment Confirmed

→ **Automatically Fired**

#### 2. Hold & Fire

The order is received but preparation does not begin until an authorized employee releases it.

Example:

Main Course:

**HELD**

Waiter selects:

**FIRE NOW**



Kitchen receives order.

#### 3. Phased / Course Firing

Different courses are fired separately.

Example:

**Starter**

Fire Now ✓

**Main Course**

Held

**Dessert**

Held

When the guest finishes the starter:

Waiter selects:

**Fire Main Course****4. Timed / Delayed Firing**

Preparation begins automatically at a configured future time.

Example:

Pre-Order:

Guest Pickup:

13:00

Preparation Time:

15 minutes

TICVAI automatically fires:

**12:45**

**Course Types**

Configurable course types may include:

Drinks

Starter

Soup

Main Course

Side

Dessert

Coffee

Custom Course

**Fire Controls**

Authorized users shall be able to:

- Fire Course
- Fire Item
- Fire Entire Order
- Hold Course
- Delay Fire
- Schedule Fire

- Cancel Fire
- Re-Fire Item

All manual firing actions shall be audited.

# SCREEN 4 — Active Order Management & Fulfilment Journey

---

## Purpose

Provide supervisors with centralized management of all active F&B orders regardless of originating channel.

## Order List

Display:

- Order #
- Outlet
- Channel
- Table / Pickup #
- Order Time
- Fire Status
- Kitchen
- Station(s)
- Current Stage
- Elapsed Time
- Target Time
- SLA
- Priority
- Payment Status
- Collection Method
- Status

## Filters

Outlet  
Kitchen  
Station  
Channel  
Course  
Fire Status  
Order Status  
Priority  
SLA  
Table  
Collection Number.

## Order Detail

Selecting an order opens the complete fulfilment journey.

Example:

**19:32:10**  
Order Submitted



**19:32:11**

Payment Confirmed



**19:32:12**

Starter Fired



**19:32:14**

Starter Routed → Cold Kitchen



**19:38:22**

Starter Ready



**19:46:00**

Main Course Fired by Waiter Ahmed



**19:46:02**

Burger → Grill

**19:46:02**

Fries → Fryer



**19:57:30**

All Main Course Items Ready



**19:58:12**

Expeditor Confirmed



**19:59:08**

Served

Every event shall record:

**Timestamp + User/System + Device + Location + Action + Status**

## SCREEN 5 — Kitchen Station Workload & Dynamic Routing

---

### Purpose

Provide live operational management of kitchen stations and intelligently route work according to configuration and real-time capacity.

## Station Cards

Example:

### GRILL STATION

Status: Active

Staff:  
4

Active Items:  
32

Queue:  
18

Average Prep:  
12.8 min

Target:  
10 min

Utilization:  
91%

SLA:  
82%

## Station Status

- Available
- Active
- Busy
- Overloaded
- Paused
- Offline
- Maintenance
- Closed

## Routing Inputs

The routing engine shall consider:

- Product
- Product Category
- Modifier
- Outlet
- Service Type
- Service Period
- Kitchen
- Station Eligibility
- Station Status
- Queue Length
- Station Capacity
- Preparation Time
- Staff Availability
- Equipment Availability

- Priority
- SLA

The additional F&B requirements specifically call for configurable kitchen routing by product, category, modifier, outlet, service type and time period, as well as priority handling for VIP/Fast Pass/SLA scenarios.

### Dynamic Routing Example

#### Grill A

Load: 97%

#### Grill B

Load: 48%

TICVAI recommends:

Route the next 10 eligible grill items to Grill B.

### Supervisor Actions

- Reassign Item
- Reassign Order
- Pause Routing
- Activate Backup Station
- Redirect Product Category
- Open / Close Station
- Temporarily Adjust Capacity

All overrides shall be logged.

## SCREEN 6 — Expeditor & Order Assembly

---

### Purpose

Provide a centralized fulfilment screen where items prepared by different kitchen stations are consolidated into a complete customer order.

Example:

#### ORDER #A1843

#### Grill

2 × Burgers  
✓ READY

#### Fryer

2 × Fries  
✓ READY

#### Beverage

2 × Drinks

✓ READY

### Dessert

1 × Cake

PREPARING

### Overall Progress

#### 3 of 4 Stations Complete

The order shall not become fully Ready until all required components for the applicable course/order have been completed.

### Expeditor Actions

- Confirm Item
- Confirm Course
- Confirm Full Order
- Mark Missing
- Hold Assembly
- Request Re-Fire
- Prioritize
- Add Note
- Request Runner
- Mark Ready
- Send to Collection

### Partial Fulfilment

The system shall support configurable rules for:

#### Serve When Complete

or

#### Serve by Course

or

#### Serve Individual Items When Ready

depending on outlet/service model.

### Handover Methods

- Waiter
- Food Runner
- Counter Pickup
- Table Delivery
- Takeaway
- Mobile Pickup
- VIP Service
- Event Delivery

## SCREEN 7 — Guest Collection, Buzzer & Digital Notification

---

### Purpose

Manage the guest collection journey once an order or course is ready.

The system shall support both traditional physical buzzers and modern digital notification channels.

### Collection Methods

#### Physical Buzzer

#### Digital Order Number

#### SMS

#### WhatsApp

#### Mobile Push Notification

#### QR Order Status

#### Digital Pickup Board

#### Waiter / Runner Notification

#### Buzzer Assignment

Example:

Order:  
#A1843

Buzzer:  
**B-042**

Status:  
Assigned

Battery:  
86%

Guest:  
Counter Pickup

When the order becomes Ready:

#### Trigger Buzzer

#### Buzzer Lifecycle

Available

→ Assigned

→ Guest Waiting

→ Alerted

→ Returned

→ Available

The matrix specifically requires the ability to assign buzzers to guests and notify them when orders are ready.

### Digital Notification

Example:

**Your order #A1843 is ready. Please collect it from Counter 2.**

### Pickup Board

Guest-facing display:

#### PREPARING

A1844

A1846

A1848

#### READY

A1841

A1843

A1845

The display should support tenant branding and multilingual presentation.

## SCREEN 8 — Exceptions, Re-Fire & Unavailable Items

---

### Purpose

Provide one operational workspace for resolving kitchen and fulfillment exceptions.

### Exception Types

- Delayed Order
- Missing Item
- Incorrect Item
- Re-Fire
- Product Unavailable
- Ingredient Unavailable
- Modifier Unavailable
- Kitchen Offline
- Station Offline
- Printer/KDS Failure
- Equipment Failure
- Allergy Escalation
- Guest Modification

- Cancellation Request
- Payment Issue
- Collection Delay

Exception Queue

| Order | Exception      | Station | Delay | Severity | Owner      | Status |
|-------|----------------|---------|-------|----------|------------|--------|
| A1843 | Re-Fire        | Grill   | 4 min | High     | Chef Omar  | Open   |
| A1848 | Ingredient OOS | Fryer   | 7 min | Critical | Supervisor | Open   |

Re-Fire Workflow

Example:

Guest reports:

Burger overcooked



Server creates:

Re-Fire Request



Reason:  
Quality Issue



Priority:  
**Recovery / Urgent**



Kitchen receives:

RE-FIRE — PRIORITY

The replacement should retain linkage to the original order and item.

Unavailable Ingredient

Kitchen employee selects:

Beef Patty — Unavailable

TICVAI identifies affected products and sends the event to Board 5's **86 / Availability Engine**.

## SCREEN 9 — SLA, Priority & Service Rules

---

### Purpose

Configure preparation targets, operational priorities and escalation rules.

### SLA Configuration

Targets may be configured by:

- Product
- Category
- Course
- Kitchen
- Station
- Outlet
- Channel
- Service Type
- Service Period
- Event
- Guest/Service Entitlement

Example:

Classic Burger:  
10 min

Coffee:  
4 min

VIP Burger:  
7 min

Event Meal:  
12 min

### Priority Levels

- Normal
- High
- VIP
- Fast Service
- Urgent
- Re-Fire / Recovery
- Event Priority
- Accessibility Service Priority

### Priority Rule Example

#### IF

Guest Entitlement = VIP

#### THEN

Priority = High

**IF**

Order = Re-Fire

**THEN**

Priority = Urgent

**IF**

Elapsed Time > SLA

**THEN**

Escalate Priority.

**SLA Warning Rules**

70%

→ Monitor

80%

→ Warning

100%

→ Delayed

120%

→ Supervisor Alert

150%

→ F&B Manager Escalation

Thresholds must be tenant-configurable.

---

## **SCREEN 10 — Kitchen Performance, AI & Operational Optimization**

---

**Purpose**

Provide real-time and shift-level kitchen intelligence and AI-assisted optimization.

**KPI Cards**

- Orders Completed
- Average Preparation Time
- Median Preparation Time
- SLA Compliance
- Delayed Orders
- Re-Fire Rate
- Items per Hour
- Station Utilization
- Peak Load

- Collection Wait Time

Station Performance

| Station  | Orders | Avg Prep | Target | SLA | Utilization |
|----------|--------|----------|--------|-----|-------------|
| Grill    | 1,284  | 13.4 min | 10     | 79% | 91%         |
| Fryer    | 940    | 8.2 min  | 8      | 94% | 73%         |
| Beverage | 1,820  | 4.1 min  | 4      | 97% | 62%         |

Firing Analysis

Measure:

- Average Hold Time
- Fire-to-Ready Time
- Course Gaps
- Delayed Fires
- Manual Fire Overrides
- Timed Fire Accuracy

Collection Analysis

Measure:

- Ready-to-Collection Time
- Buzzer Response Time
- Digital Notification Response
- Orders Waiting Too Long After Ready

AI KITCHEN OPERATIONS ASSISTANT

Authorized supervisors should be able to ask:

Why is Grill Station delayed?

Which orders are likely to miss SLA?

Which station should handle the next burger orders?

Why did preparation time increase after 7 PM?

Should I activate the backup kitchen?

Which items are creating bottlenecks?

Which held courses should be fired soon?

Example AI Response

**Grill Station is operating at 96% capacity.**

27 grill items are currently active and average preparation time has increased from 9.8 to 13.7 minutes.

**Recommended action:** Route eligible burger orders to Grill B for the next 35 minutes.

Actions:

**Simulate**

**Approve Rerouting**

**Ignore**

## **AI — Predictive Firing**

For timed dining experiences, AI may recommend when to fire a course.

Example:

Table 18

Main Course:  
Held

Starter status:  
75% complete

Historical service gap:  
8 minutes

Main Course preparation:  
14 minutes

TICVAI recommends:

**Fire Main Course in approximately 4 minutes** to maintain the target service interval.

The waiter or supervisor remains in control unless automated firing has been explicitly enabled.

## **AI — Delay Prevention**

TICVAI should identify orders likely to become late before SLA is breached.

Example:

**ORDER A1843**

Probability of missing SLA:

**84%**

Reason:

Grill Load:  
96%

Queue:  
18 items

Current Prep Variance:  
+31%

Recommendation:

### **Reroute Burger → Grill B**

This supports proactive rather than reactive kitchen management.

## **AI — Kitchen Load Forecasting**

Because TICVAI also manages ticketing, reservations, events and visitor demand, Board 3 should consume forecasts from the broader platform.

Example:

### **18:00–20:00**

Expected Park Attendance:  
8,300

Expected F&B Orders:  
2,650

Forecast:

Grill:  
108%

Fryer:  
94%

Beverage:  
81%

Recommendation:

- Open Backup Grill at 17:30
- Add 2 Grill Staff
- Add 1 Expeditor
- Prepare peak service mode

Staff recommendations should integrate with TICVAI Resource Management.

## **CROSS-BOARD REQUIREMENTS**

### **Unified Order Engine**

POS, QR, mobile, kiosk, dine-in and other channels must **not have independent kitchen workflows**.

All orders shall normalize into:

## Unified F&B Order

- Fulfilment Rules
- Firing
- Routing
- Kitchen
- Expeditor
- Collection/Service.

## Real-Time Synchronization

Order statuses shall synchronize across:

- POS
- KDS
- Waiter interface
- Expeditor
- Guest App
- QR Ordering
- Collection Display
- Employee App

The additional requirements call for synchronized operational statuses such as New, In Preparation, Ready, Served and Completed.

## Offline Resilience

Kitchen operations shall support appropriate offline resilience including:

- Local queue retention
- Order caching
- Automatic synchronization
- Duplicate prevention
- Reconnection handling
- Device health monitoring

No confirmed kitchen order should disappear because of a temporary connectivity interruption.

## Multi-Kitchen / Multi-Outlet

The architecture shall support:

### One Outlet → Multiple Kitchens

and

### One Kitchen → Multiple Outlets

and

### **One Order → Multiple Stations**

This is mandatory for enterprise parks, attractions, resorts and multi-outlet environments.

### **Audit Trail**

The system shall audit:

- Fire
- Hold
- Scheduled Fire
- Re-Fire
- Routing
- Reassignment
- Priority Change
- Cancellation
- Station Closure
- Manual Override
- SLA Change
- Buzzer Assignment
- Ready Status
- Collection Confirmation

Audit records shall include:

**User + Timestamp + Device + Location + Previous Value/Status + New Value/Status + Reason**

### **Employee App Integration**

Board 3 shall integrate with the TICVAI Employee App.

Examples of alerts:

Grill Station overloaded.

VIP Order #1843 approaching SLA.

KDS-04 offline.

Buzzer B-042 has not been returned.

9 orders waiting for collection.

Main Course for Table 18 ready.

Authorized supervisors may:

- Approve Rerouting
- Fire Course
- Activate Backup Station
- Acknowledge Alert
- Assign Staff

- Report Equipment Issue
- Escalate Incident

Board 3 — Final 10 Screens

| #  | Screen                                             |
|----|----------------------------------------------------|
| 1  | Kitchen Operations Command Center                  |
| 2  | Kitchen Display System (KDS)                       |
| 3  | Order Firing & Course Management                   |
| 4  | Active Order Management & Fulfilment Journey       |
| 5  | Kitchen Station Workload & Dynamic Routing         |
| 6  | Expeditor & Order Assembly                         |
| 7  | Guest Collection, Buzzer & Digital Notification    |
| 8  | Exceptions, Re-Fire & Unavailable Items            |
| 9  | SLA, Priority & Service Rules                      |
| 10 | Kitchen Performance, AI & Operational Optimization |

Complete Board 3 Operational Flow

Order Confirmed



Unified F&B Order Engine



Determine:

Immediate Fire / Hold / Course Fire / Timed Fire



Routing Engine



Kitchen + Preparation Station



KDS



Accept + Prepare



Real-Time SLA Monitoring



Station Items Ready



**Expeditor**



Course / Order Assembled



**Ready**



Determine Fulfilment:

**Waiter / Runner / Buzzer / Digital Notification / Pickup Board**



**Served / Collected**



**Completed**



Analytics + Inventory + CRM + Finance updated.

### **Architectural Requirement**

The solution shall use a common **F&B Fulfilment Engine** rather than creating separate backend processes for different channels or outlet types.

The architecture shall follow:

#### **Order Source**

POS / QR / Mobile / Kiosk / B2C / Table / VIP / Event



#### **Unified F&B Order**



#### **Firing Engine**



#### **Routing & Priority Engine**



## **Kitchen / KDS**



## **Preparation Stations**



## **Expeditor**



## **Collection / Service Engine**



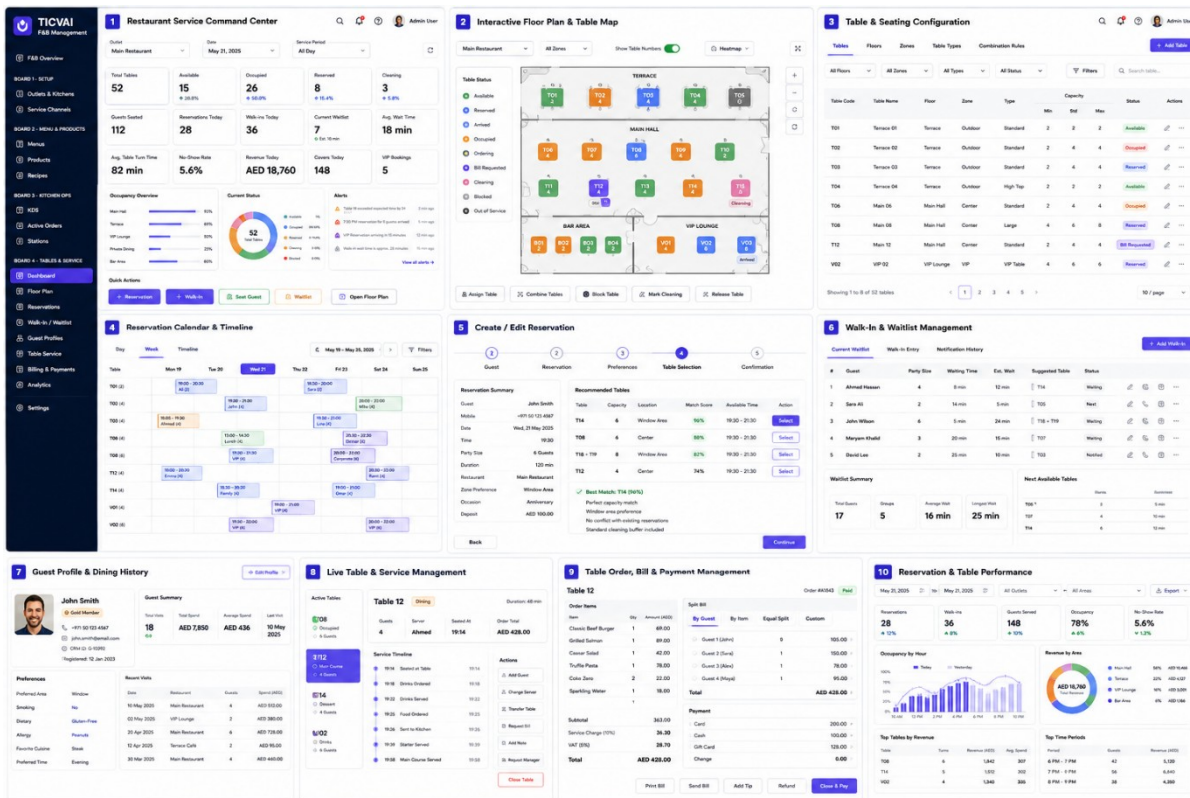
## **Completion**



## **Analytics & AI**

Hold & Fire, phased courses, timed preparation, priority, buzzer management and guest notifications should therefore be configurable capabilities of the same fulfilment engine—not separate systems.

# TICVAI F&B — BOARD 4



## Tables, Reservations & Service Management — Scope of Work

### 1. Purpose

The module shall provide centralized management of:

**Restaurant → Floor/Zone → Table → Capacity → Reservation → Guest → Seating → Order → Service → Payment → Turnover**

It shall support:

- Full-service restaurants
- Cafés
- VIP lounges
- Hospitality areas
- Event dining
- Walk-in restaurants
- Reservation-only restaurants
- Mixed reservation/walk-in operations

The system must integrate with TICVAI CRM, POS, F&B ordering, loyalty/membership, ticketing, Resource Management, Employee App and Finance.

## Screen 1 — Restaurant Service Command Center

This is the **live operational dashboard** for the restaurant manager.

### KPI cards

- Total Tables

- Available Tables
- Occupied Tables
- Reserved Tables
- Tables Cleaning
- Guests Seated
- Reservations Today
- Walk-ins Today
- Current Waitlist
- Average Wait Time
- Average Table Turn Time
- No-Show Rate

Live restaurant status

Example:

| Area      | Tables | Occupied | Reserved | Available | Guests |
|-----------|--------|----------|----------|-----------|--------|
| Main Hall | 28     | 18       | 4        | 6         | 71     |
| Terrace   | 16     | 11       | 2        | 3         | 39     |
| VIP       | 8      | 4        | 3        | 1         | 22     |

Alerts

Table 18 has exceeded expected dining time by 24 minutes.

7:30 PM reservation for 8 guests has arrived.

VIP reservation arriving in 15 minutes.

Current walk-in wait time is approximately 28 minutes.

Quick actions

+ Reservation | + Walk-In | Seat Guest | Open Floor Plan | Waitlist | Live Service

Screen 2 — Floor Plan & Table Map

This should be one of the strongest visual screens in Board 4.

The system shall provide an interactive **2D restaurant floor plan**.

Example

MAIN RESTAURANT

Window Area

[T01]

4

Available

[T02]

4

Occupied

[T03]

6

Reserved

19:30

BAR

[T04] [T05] [T06]

MAIN HALL

[T07] [T08] [T09]

VIP SECTION

[V01] [V02] [V03]

Tables should display live status.

### **Statuses**

**Available**

**Reserved**

**Guest Arrived**

**Occupied**

**Ordering**

**Order Preparing**

**Dining**

**Bill Requested**

**Payment Pending**

**Cleaning**

**Blocked**

**Out of Service**

### **Floor plan actions**

Authorized staff should be able to:

- Drag tables
- Move tables
- Combine tables
- Split tables
- Block tables
- Assign reservation
- Seat walk-in
- Transfer guest
- Open order
- View bill
- Mark cleaning
- Release table

### **Heatmap mode**

Managers should be able to switch the floor plan into:

**Occupancy Heatmap**

or

**Revenue Heatmap**

or

### Turnover Heatmap

This becomes very useful operationally.

## Screen 3 — Table & Seating Configuration

---

Screen 2 operates the tables.

Screen 3 **configures** them.

### Table master

Each table shall contain:

- Table Name
- Table Code
- Restaurant
- Floor
- Zone
- Minimum Capacity
- Standard Capacity
- Maximum Capacity
- Table Shape
- Table Type
- Accessible
- Outdoor / Indoor
- Smoking / Non-Smoking where applicable
- VIP
- Status

### Table types

Examples:

Standard

High Top

Booth

Bar Seat

VIP Table

Private Dining

Terrace

Family Table

### Combination rules

Example:

T10 + T11

Can combine to create:

#### 8-seat table

T11 + T12

Can combine.

But:

T10 + T15

Cannot combine.

This allows the reservation engine to understand physical seating possibilities.

## Screen 4 — Reservation Calendar & Timeline

This is the primary reservation-planning interface.

The screen should provide:

**Day | Week | Timeline**

### Timeline example

| Time  | T01     | T02   | T03   | T04     |
|-------|---------|-------|-------|---------|
| 18:00 | Smith 4 | —     | —     | Ahmed 2 |
| 18:30 | Smith 4 | —     | —     | Ahmed 2 |
| 19:00 | —       | Ali 6 | —     | —       |
| 19:30 | John 4  | Ali 6 | VIP 8 | —       |
| 20:00 | John 4  | —     | VIP 8 | Sara 2  |

### Filters

- Restaurant
- Zone
- Table
- Party Size
- Reservation Status
- Guest Type
- VIP
- Membership
- Channel
- Service Period

### Reservation statuses

**Pending**

**Confirmed**

**Arriving Soon**

**Arrived**

**Seated**

**Completed**

**Cancelled**

**No-Show**

## Screen 5 — Create / Edit Reservation

This is the detailed reservation workflow.

### Step 1 — Guest

Search existing guest by:

- Mobile

- Email
- Membership
- Ticket
- Name

Or:

### + Create Guest

## Step 2 — Reservation

Fields:

- Date
- Time
- Party Size
- Restaurant
- Zone Preference
- Table Preference
- Duration
- Occasion

### Occasion

Birthday

Anniversary

Business Dinner

VIP Visit

Family

Corporate Event

Other

## Step 3 — Guest preferences

- Indoor / Outdoor
- Smoking / Non-Smoking
- Accessibility requirement
- High chair
- Wheelchair space
- Dietary requirements
- Allergies
- Special request

## Step 4 — Table suggestion

TICVAI should recommend:

### Best Match

Table T14

Capacity: 6

Party: 5

Available: 19:30–21:30

Preference Match: 96%

## Step 5 — Confirmation

Send confirmation via:

- Email
- SMS

- WhatsApp
- Mobile App

Optional:

Add to Wallet / Calendar.

## Screen 6 — Walk-In & Waitlist Management

This is essential for restaurants that do not operate exclusively by reservation.

### Walk-in entry

Capture:

- Guest Name
- Mobile
- Party Size
- Arrival Time
- Preferred Area
- Accessibility
- Special Requirement

### Waitlist

| Guest | Party | Waiting | Est. Wait | Suggested Table | Status  |
|-------|-------|---------|-----------|-----------------|---------|
| Ahmed | 4     | 8 min   | 12 min    | T14             | Waiting |
| Sara  | 2     | 14 min  | 5 min     | T05             | Next    |
| John  | 6     | 5 min   | 24 min    | T18+T19         | Waiting |

### Guest communication

Guest receives:

You're currently #3 on the waitlist.

Later:

Your table will be ready in approximately 10 minutes.

Then:

Your table is ready. Please proceed to the restaurant host.

### Waitlist statuses

Waiting

Notified

Arrived

Seated

Cancelled

Left

## Screen 7 — Guest Profile & Dining History

This is where F&B connects properly to the **TICVAI CRM**.

It should not create a separate restaurant customer database.

### Guest profile

Show:

**John Smith**

CRM ID: G-10392

Membership:

Gold

Visits:

18

F&amp;B Spend:

AED 7,850

Average Spend:

AED 436

No-Shows:

1

**Preferences**

Preferred:

Terrace

Favorite:

Steak

Drink:

Sparkling Water

Dietary:

Gluten-Free

Allergy:

Peanuts

Preferred seating:

Window

**History**

Show previous:

- Reservations
- Orders
- Restaurants
- Spend
- Promotions
- Complaints
- Feedback

**Important architecture**

This data should come from the **shared TICVAI Guest Profile / CRM**.

F&B simply extends it with dining preferences.

---

## Screen 8 — Live Table & Service Management

---

This screen manages the restaurant **after guests have been seated**.

**Table detail**

Example:

**TABLE 12**

Guests: 4

Server: Ahmed

Seated: 19:14

Dining Duration: 48 min

Current Stage:

### Main Course

Order:

AED 428

### Service timeline

19:14 — Seated

19:18 — Drinks Ordered

19:22 — Drinks Served

19:25 — Food Ordered

19:26 — Sent to Kitchen

19:39 — Starter Served

19:58 — Main Course Served

20:14 — Dessert Offered

### Actions

- Add Guest
- Remove Guest
- Change Server
- Transfer Table
- Combine Table
- Move Order
- Request Kitchen
- Add Note
- Request Manager
- Request Bill
- Close Table

## Screen 9 — Table Order, Bill & Payment Management

---

This connects Board 4 directly with POS and Finance.

### Table order

Example:

#### Table 12

Guest 1:

Burger

Coke

Guest 2:

Steak

Water

Guest 3:

Salad

Juice

Guest 4:

Pizza

Coffee

### **Split bill**

The system should support:

#### **Split by Guest**

#### **Split by Item**

#### **Split Equally**

#### **Custom Amount**

### **Example**

Total:

AED 428

Split:

Guest 1 → AED 105

Guest 2 → AED 150

Guest 3 → AED 78

Guest 4 → AED 95

### **Payment methods**

Cash

Card

Mobile Wallet

Gift Card

Voucher

Membership Credit

Room Charge, where integrated

Corporate Account

### **Mixed tender**

Example:

AED 200 Card

AED 100 Cash

AED 128 Gift Card

### **Service charge / gratuity**

Support:

- Fixed
- Percentage
- Optional
- Automatic
- Manual

Refund, discount, complimentary and override actions must use TICVAI's shared approval and audit framework.

---

## **Screen 10 — Reservation & Table Performance**

---

This page provides operational analytics for restaurant managers.

### **KPIs**

- Reservations
- Walk-Ins

- Guests Served
- Covers
- Occupancy %
- Average Wait Time
- Average Dining Time
- Table Turnover
- Revenue per Table
- Revenue per Cover
- No-Show %
- Cancellation %

**Table utilization**

Example:

| Table | Occupancy | Turns | Revenue   | Avg. Spend |
|-------|-----------|-------|-----------|------------|
| T01   | 84%       | 5     | AED 3,420 | AED 684    |
| T02   | 72%       | 4     | AED 2,860 | AED 715    |
| T03   | 91%       | 6     | AED 4,210 | AED 702    |

**Analytics dimensions**

Analyze by:

Restaurant  
Zone  
Table  
Hour  
Day  
Service Period  
Party Size  
Guest Segment  
Reservation Channel

**Reservation Capacity Engine**

I strongly recommend adding this capability.

TICVAI shouldn't simply ask:

Is Table 10 free at 7 PM?

It should calculate:

**Table Capacity + Party Size + Reservation Duration + Combination Rules + Current Occupancy + Future Reservations + Turnaround Buffer**

Example:

Reservation:

6 guests  
7:30 PM  
Expected duration: 120 min

TICVAI determines:

T12 capacity = 4

T14 capacity = 6 ✓

T18 + T19 = 8 ✓

TICVAI recommends:

**T14 — Best Match**

rather than wasting an eight-seat combined table.

**Configurable Reservation Duration**

Duration should depend on:

Party size

Restaurant

Service period

Day

Guest type

Example:

2 guests → 90 min

4 guests → 120 min

6–8 guests → 150 min

VIP → 180 min

The venue must be able to override these rules.

**Turnaround / Cleaning Buffer**

The reservation engine must consider cleaning time.

Example:

Reservation:

19:00–21:00

Cleaning buffer:

15 minutes

Next available:

**21:15**

Not 21:00.

This also connects later to Employee/Resource Management if cleaning tasks need assignment.

**Deposit & Reservation Guarantee**

I recommend including optional reservation deposits.

Restaurants may configure:

**No Deposit**

or

**Fixed Deposit**

AED 100

or

**Per Guest**

AED 50 × guest

or

**Percentage**

20%

or

**Card Guarantee**

Rules can depend on:

Restaurant

Party Size

VIP Area

Date

Event

Peak Period

Example:

New Year's Eve

Reservation deposit:

### **AED 250 per guest**

Cancellation policy:

Free cancellation until 48 hours before.

This should use TICVAI's shared payment engine.

### **No-Show & Cancellation Rules**

The system should support configurable:

- Cancellation Window
- Cancellation Fee
- No-Show Fee
- Deposit Forfeiture
- Refund Rules
- Grace Period

Example:

Reservation: 19:30

Grace Period: 15 min

At 19:45:

#### **Late**

At 20:00:

#### **No-Show**

Rules remain configurable.

### **Ticketing + F&B Reservation Integration**

This is particularly important for TICVAI.

Because TICVAI is primarily an attraction/ticketing platform, restaurants should be able to link reservations to:

#### **Ticket**

#### **Membership**

#### **Experience Booking**

#### **Event**

#### **VIP Package**

Example:

Guest purchases:

#### **Theme Park VIP Package**

Includes:

Park Admission

Fast Track

Lunch Reservation at 13:00

VIP Lounge Access

The F&B reservation should be automatically created from the package.

This is a significant advantage compared with a standalone restaurant POS.

### **Membership & Loyalty Integration**

Example:

Gold Member:

Priority Reservation ✓

Priority Waitlist ✓

10% F&B Discount ✓

Complimentary Dessert ✓

VIP Table Access ✓

The reservation engine should recognize the guest's membership automatically.

### **AI — SMART TABLE ASSIGNMENT**

AI should recommend the optimal table rather than simply showing availability.

It should consider:

- Party Size
- Table Capacity
- Guest Preference
- Accessibility
- VIP Status
- Reservation Duration
- Upcoming Reservations
- Current Table Status
- Historical Dining Time
- Table Combination
- Restaurant Load

Example:

#### **Recommended: Table 14 — 96% Match**

Reason:

Capacity fits party exactly.

Guest previously requested window seating.

No conflicting reservation until 22:00.

### **AI — WAIT TIME PREDICTION**

Instead of:

Estimated wait: 30 minutes

TICVAI should predict wait time based on live operations.

Example:

#### **Predicted wait: 18–24 minutes**

Based on:

4 tables currently requesting bills.

2 tables currently cleaning.

Historical turnover.

Current kitchen/service load.

### AI — NO-SHOW PREDICTION

For appropriate business use, TICVAI may flag reservations with elevated operational no-show risk based on non-sensitive booking behavior such as:

- Reservation lead time
- Confirmation status
- Historical reservation attendance
- Deposit status

Example:

Reservation #R1842 has an elevated no-show risk.

Recommended operational action:

#### **Send confirmation reminder.**

The system should avoid using sensitive personal characteristics for this prediction.

### AI — DEMAND & CAPACITY FORECASTING

AI should forecast:

Expected covers tonight: **486**

Peak period:

**19:00–20:30**

Expected occupancy:

**96%**

Predicted walk-in demand:

**72 guests**

Recommendation:

Keep Tables T12–T16 available for walk-ins between 18:30 and 19:30.

This can use:

Ticket sales

Event attendance

Historical restaurant data

Reservations

Seasonality

Day/time.

### Employee App Integration

Restaurant staff should be able to receive operational tasks on the TICVAI Employee App.

Examples:

VIP guest arrived — Reservation R-1842.

Table 14 requires cleaning.

Table 8 requested bill.

Table 12 waiting 8 minutes for service.

Party of 6 ready to be seated.

Staff can acknowledge and complete tasks.

### Approval & Audit Controls

Sensitive actions must follow TICVAI's shared approval framework.

Examples:

### Complimentary Item

→ Supervisor approval

### Large Discount

→ Manager approval

### Reservation Deposit Refund

→ Finance / Manager approval

### VIP Override

→ Authorized role

### Manual Table Block

→ Restaurant Manager

All significant actions must capture:

User

Date/Time

Action

Reason

Old Value

New Value

Device

Approval

## Board 4 — Final 10 Screens

| #  | Screen                                 |
|----|----------------------------------------|
| 1  | Restaurant Service Command Center      |
| 2  | Interactive Floor Plan & Table Map     |
| 3  | Table & Seating Configuration          |
| 4  | Reservation Calendar & Timeline        |
| 5  | Create / Edit Reservation              |
| 6  | Walk-In & Waitlist Management          |
| 7  | Guest Profile & Dining History         |
| 8  | Live Table & Service Management        |
| 9  | Table Order, Bill & Payment Management |
| 10 | Reservation & Table Performance        |

## Complete Board 4 workflow

### Reservation / Walk-In

→ Guest Identified

→ Availability & Capacity Check

→ Table Recommended

→ Reservation Confirmed / Added to Waitlist

→ Guest Arrives

→ Host Checks In

→ Table Assigned

→ Guest Seated

→ Server Assigned

- Order Created
- **Board 3 Kitchen Fulfilment**
- Food Served
- Dining
- Bill Requested
- Split / Payment
- Table Closed
- Cleaning
- **Available Again**
- CRM + Finance + Analytics Updated

### **Architectural requirement**

Development team should **not create separate reservation engines** for restaurants, VIP lounges, dining packages and event dining.

There should be a common **Reservation & Capacity Engine**:

**Bookable Entity → Capacity → Availability → Time Slot → Guest → Rules → Reservation → Payment/Deposit → Status**

The F&B layer then adds:

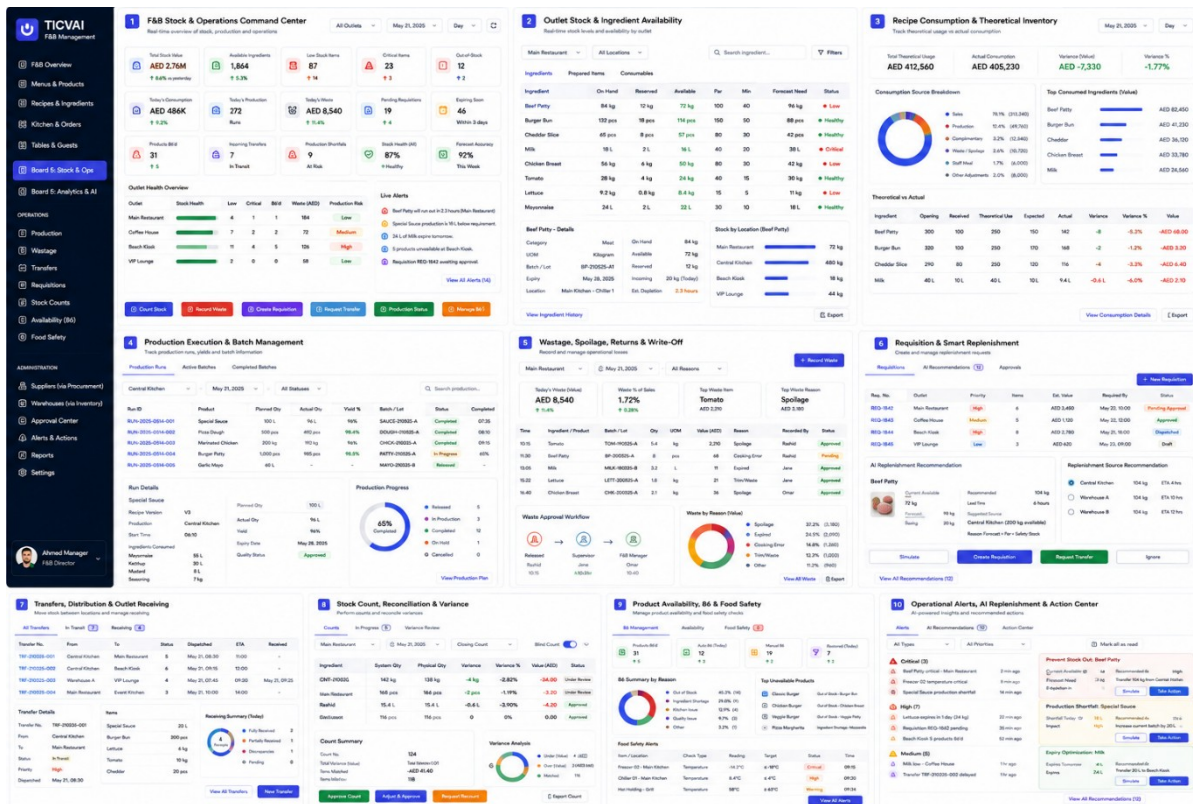
**Table → Party Size → Seating Preference → Dining Duration → Waitlist → Service → Table Turnover**

This also aligns very well with the common Resource Engine architecture: restaurant tables can behave as bookable resources while the F&B module provides the specialized restaurant rules.

That approach will make it much easier later for TICVAI to support restaurant reservations bundled with **tickets, events, memberships, VIP packages and experiences** without creating separate booking logic for every product type.

# TICVAI F&B — BOARD 5

## F&B Stock, Production, Replenishment & Operational Control — Scope of Work



### 1. Purpose

The F&B Stock, Production, Replenishment & Operational Control Board shall provide real-time management of ingredients, prepared items, consumables, production stock, outlet stock, wastage, stock counts, replenishment, internal transfers, availability, and operational food-safety controls across all Food & Beverage operations.

The Board shall connect F&B sales and production activities with TICVAI's shared Inventory & Procurement architecture without creating a separate inventory system inside the F&B module.

The core operational lifecycle shall be:

**Sale / Forecast → Recipe Consumption → Theoretical Stock → Actual Stock → Variance → Production Requirement → Replenishment → Outlet Availability**

The module shall support:

- Restaurants
- Cafés
- Kiosks
- Food Trucks
- Bars
- VIP Lounges
- Event Catering
- Central Kitchens
- Commissaries
- Temporary Event Outlets

- Mobile F&B Operations

The Board shall support both **outlet-level operations** and **central-kitchen production flows**.

## SCREEN 1 — F&B Stock & Operations Command Center

### Purpose

Provide F&B managers with a centralized real-time view of stock health, consumption, production, wastage, replenishment and operational availability across all outlets.

### Global Filters

- Tenant
- Company
- Park
- Venue
- Outlet
- Central Kitchen
- Stock Location
- Date
- Service Period

### KPI Cards

- Total F&B Stock Value
- Available Ingredients
- Low Stock Items
- Critical Stock Items
- Out-of-Stock Items
- Products 86'd
- Today's Consumption
- Today's Production
- Today's Waste
- Waste Value
- Pending Requisitions
- Incoming Transfers
- Expiring Soon
- Production Shortfalls

### Outlet Health

| Outlet          | Stock Health | Low | Critical | 86'd | Waste   | Production Risk |
|-----------------|--------------|-----|----------|------|---------|-----------------|
| Main Restaurant | 92%          | 4   | 1        | 1    | AED 184 | Low             |
| Coffee House    | 88%          | 7   | 2        | 2    | AED 72  | Medium          |
| Beach Kiosk     | 73%          | 11  | 4        | 5    | AED 126 | High            |
| VIP Lounge      | 96%          | 2   | 0        | 0    | AED 58  | Low             |

### Live Alerts

Examples:

Beef Patty stock at Main Restaurant is forecast to run out within 2.3 hours.

Central Kitchen production of Special Sauce is 18 litres below today's requirement.

24 litres of milk expire tomorrow.

Beach Kiosk has five products unavailable due to ingredient shortages.

Main Restaurant requisition REQ-1842 is awaiting approval.

Quick Actions

Count Stock | Record Waste | Create Requisition | Request Transfer | Production Status | Manage 86 | View Alerts

SCREEN 2 — Outlet Stock & Ingredient Availability

Purpose

Provide real-time visibility of ingredients, prepared items and consumables available at each F&B operational location.

Stock Dimensions

The system shall distinguish between:

- On Hand
- Reserved
- Available
- In Preparation
- In Transit
- Incoming
- Blocked
- Damaged
- Expiring
- Unavailable

Stock Table

| Ingredient | On Hand | Reserved | Available | Par  | Min  | Forecast Need | Status   |
|------------|---------|----------|-----------|------|------|---------------|----------|
| Beef Patty | 84      | 12       | 72        | 100  | 40   | 96            | Low      |
| Burger Bun | 132     | 18       | 114       | 150  | 50   | 88            | Healthy  |
| Cheddar    | 65      | 8        | 57        | 80   | 30   | 42            | Healthy  |
| Milk       | 18 L    | 2 L      | 16 L      | 40 L | 20 L | 38 L          | Critical |

Stock Drill-Down

Selecting an ingredient shall display:

- Current Stock
- Available Stock
- Reserved Stock
- Batch / Lot
- Expiry Date
- Storage Location
- Last Receipt
- Last Issue
- Last Transfer
- Today's Consumption
- Historical Consumption
- Forecast Consumption
- Current Par Level
- Reorder Level
- Expected Depletion Time

**Multi-Location Visibility**

Authorized users shall be able to see the same ingredient across locations.

Example:

**Beef Patty**

Main Restaurant — 72 available

Beach Kiosk — 18 available

VIP Lounge — 44 available

Central Kitchen — 480 available

TICVAI should therefore be able to recommend internal redistribution before external procurement.

**SCREEN 3 — Recipe Consumption & Theoretical Inventory**

---

**Purpose**

Automatically convert F&B sales and production activities into ingredient-level consumption using the approved recipes configured in Board 2.

**Consumption Flow**

**Product Sold**



**Recipe Identified**



**Variant / Modifier Applied**



Ingredient Quantities Calculated



Inventory Consumption Event



Shared Inventory Ledger Updated

Example

250 Classic Burgers sold.

Approved recipe requires:

- 1 Beef Patty
- 1 Burger Bun
- 1 Cheese Slice
- 20g Lettuce
- 30g Tomato
- 20ml Sauce

Theoretical consumption:

- 250 Beef Patties
- 250 Buns
- 250 Cheese Slices
- 5 kg Lettuce
- 7.5 kg Tomato
- 5 L Sauce

Theoretical vs Actual

| Ingredient | Opening | Received | Theoretical Use | Expected | Actual | Variance |
|------------|---------|----------|-----------------|----------|--------|----------|
| Beef Patty | 300     | 100      | 250             | 150      | 142    | -8       |
| Burger Bun | 320     | 100      | 250             | 170      | 168    | -2       |
| Cheese     | 290     | 80       | 250             | 120      | 116    | -4       |

Consumption Sources

The system shall distinguish between:

- Sale
- Modifier
- Combo Component
- Production
- Sample
- Staff Meal
- Complimentary Item
- Waste
- Spoilage
- Re-Fire
- Manual Adjustment

- Transfer

This is essential for explaining stock variance rather than simply reporting it.

## SCREEN 4 — Production Execution & Batch Management

---

### Purpose

Execute the production plans and production sheets created from Board 2 and manage actual preparation quantities, yields, batches and completed production.

Board 2 determines:

### What should be produced.

Board 5 records:

### What was actually produced.

### Production Run

Example:

#### **RUN-2026-0814-042**

Production Item:

Special Sauce

Central Kitchen

Planned Quantity:

100 L

Actual Quantity:

96 L

Yield:

96%

Batch:

SAUCE-140826-A

Production Start:

06:10

Completed:

07:35

Expiry:

17 Aug 2026

### Production Lifecycle

**Planned**

- Released
- Ingredients Allocated
- In Production
- Quality Check
- Completed
- Released to Stock
- Distributed / Consumed

**Production Consumption**

When production begins, TICVAI shall consume the raw ingredients according to the approved production recipe.

Example:

100 L Special Sauce requires:

Mayonnaise — 55 L  
Ketchup — 30 L  
Mustard — 8 L  
Seasoning — 7 kg

Actual production:

96 L

The system calculates:

**Expected Yield vs Actual Yield**

and identifies production variance.

**Production Batch Information**

Capture:

- Batch Number
- Production Item
- Recipe Version
- Planned Quantity
- Actual Quantity
- UOM
- Yield
- Production Date
- Expiry
- Production Kitchen
- Production Team
- Quality Status
- Storage Location

## SCREEN 5 — Wastage, Spoilage, Returns & Write-Off

---

### Purpose

Provide controlled recording and approval of F&B operational losses.

### Waste Categories

- Expired
- Spoiled
- Overproduction
- Preparation Waste
- Trim / Yield Loss
- Cooking Error
- Burned
- Dropped
- Guest Return
- Re-Fire
- Quality Failure
- Temperature Failure
- Damaged Packaging
- Sample / Tasting
- Staff Meal
- Other

### Waste Entry

Capture:

- Outlet / Kitchen
- Ingredient / Product
- Batch
- Quantity
- UOM
- Unit Cost
- Total Cost
- Reason
- Employee
- Date / Time
- Photo, optional
- Notes

### Example

#### Beef Patty

Quantity:  
8 pcs

Cost:  
AED 68

Reason:  
Cooking Error

Related Station:  
Grill

Related Order:  
Optional

### Approval Workflow

Example:

Waste ≤ AED 100  
→ Supervisor

AED 101–500  
→ F&B Manager

Above AED 500  
→ F&B Director / Finance

Thresholds shall be configurable.

### Waste Analytics

The screen should immediately show:

Today's Waste

Waste %

Waste vs Target

Top Waste Item

Top Waste Reason

Waste Cost by Outlet

## SCREEN 6 — Requisition & Smart Replenishment

---

### Purpose

Allow outlets and production kitchens to request replenishment from warehouses, central kitchens or other approved stock locations.

### Requisition Information

- Requisition Number
- Requesting Location
- Supply Location
- Required Date / Time
- Priority
- Item
- Current Available
- Par Level

- Minimum Level
- Forecast Requirement
- Requested Quantity
- Recommended Quantity
- UOM
- Reason
- Requested By

### Example

#### Milk

Current:  
16 L

Par:  
40 L

Forecast Requirement:  
52 L

Incoming:  
10 L

Recommended Replenishment:

**36 L**

### Requisition Lifecycle

#### Draft

→ Submitted

→ Approval

→ Approved

→ Picking

→ Dispatched

→ In Transit

→ Received

→ Closed

### Automatic Requisition Recommendation

The system shall be able to create a suggested requisition based on:

- Current Available Stock
- Par Level
- Minimum Stock
- Forecast Demand

- Reservations
- Expected Attendance
- Production Plan
- Incoming Stock
- Safety Stock
- Lead Time

The system should initially generate a **recommendation**, with authorized users able to approve, modify or reject it.

## SCREEN 7 — Transfers, Distribution & Outlet Receiving

---

### Purpose

Manage movement of F&B inventory and prepared production between operational locations.

### Supported Transfers

**Warehouse → Outlet**

**Central Kitchen → Outlet**

**Outlet → Outlet**

**Outlet → Event Location**

**Outlet → Central Kitchen**

**Temporary Outlet → Main Outlet**

### Central Kitchen Distribution

Example:

Production Batch:

#### SAUCE-140826-A

Available:

96 L

Distribution:

Main Restaurant → 40 L

Beach Kiosk → 20 L

VIP Lounge → 15 L

Event Kitchen → 21 L

### Transfer Status

- Requested
- Approved
- Picking
- Packed
- Dispatched
- In Transit
- Partially Received
- Received
- Discrepancy
- Rejected
- Closed

**Receiving**

Outlet employee verifies:

**Sent**

100 units

**Received**

98

**Damaged**

2

TICVAI creates:

**Receiving discrepancy — 2 units**

**Transfer Traceability**

The system shall retain:

Source Location

Destination

Batch

Quantity

Sender

Receiver

Dispatch Time

Receipt Time

Discrepancy

Reason.

## SCREEN 8 — Stock Count, Reconciliation & Variance

### Purpose

Support operational stock counts and reconciliation between physical and system inventory.

### Count Types

- Opening Count
- Closing Count
- Shift Count
- Spot Count
- Cycle Count
- Full Count
- Investigation Count

### Count Screen

| Ingredient | System | Physical | Variance | Variance % | Value   |
|------------|--------|----------|----------|------------|---------|
| Beef Patty | 142    | 138      | -4       | -2.8%      | -AED 34 |
| Burger Bun | 168    | 166      | -2       | -1.2%      | -AED 3  |
| Milk       | 16 L   | 15.4 L   | -0.6 L   | -3.8%      | -AED 4  |

### Blind Count

The solution shall support:

#### Blind Count = ON

When enabled, employees performing the count shall not see the expected system quantity before submitting their physical count.

### Variance Workflow

#### Count Submitted



Variance Detected



Tolerance Check



If within tolerance:

Auto-process according to policy

If outside tolerance:

#### Supervisor Review



Recount / Explanation



Approval



Inventory Adjustment

### **Variance Investigation**

TICVAI should help identify possible sources:

Sales Consumption

Unrecorded Waste

Production Variance

Transfer Difference

Receiving Difference

Recipe Change

Re-Fire

Manual Adjustment.

## **SCREEN 9 — Product Availability, 86 & Operational Food Safety**

---

### **Purpose**

Provide centralized operational control over F&B product availability while also surfacing food-safety conditions that may affect whether products can be sold.

### **86 Management**

A product, variant, modifier or ingredient may become unavailable because of:

- Out of Stock
- Ingredient Shortage
- Kitchen Issue
- Equipment Failure
- Quality Issue
- Food Safety
- Temporary Management Decision

### **Automatic 86**

Example:

**Burger Bun = 0**

Affected:

Classic Burger  
Chicken Burger  
Veggie Burger

TICVAI identifies:

**3 products affected by Burger Bun shortage.**

Recommended:

**Mark Unavailable****Channel Synchronization**

When confirmed, availability shall update across applicable:

- POS
- Mobile POS
- Kiosk
- QR Ordering
- Mobile App
- B2C
- Digital Menu

**Outlet-Level Availability**

Classic Burger:

Main Restaurant — Available

Beach Kiosk — **Unavailable**

VIP Lounge — Available

**Automatic Restoration**

When stock arrives:

Burger Bun:

0 → 120

TICVAI identifies:

Stock restored. Three products may now be made available again.

Action:

**Restore Products**

## Food Safety Operational Controls

The system should support configurable food-safety checks including:

- Refrigerator Temperature
- Freezer Temperature
- Hot Holding
- Cold Holding
- Receiving Temperature
- Cooking Temperature
- Cleaning Check
- Storage Check

### Example

#### Freezer 02

Target:  
 $\leq -18^{\circ}\text{C}$

Reading:  
 $-14^{\circ}\text{C}$

Status:

#### Critical

TICVAI may automatically identify potentially affected stock batches and place them into:

#### Quality Hold

until an authorized employee resolves the issue.

#### Future IoT Integration

The architecture should allow future temperature sensors and IoT devices to submit readings automatically.

Manual readings must also be supported.

---

## SCREEN 10 — Operational Alerts, AI Replenishment & Action Center

---

### Purpose

Provide one centralized workspace for identifying F&B stock and production risks and converting them into controlled operational actions.

### Alert Categories

#### Stock

- Low Stock
- Critical Stock
- Out of Stock
- Overstock

**Production**

- Production Behind Plan
- Yield Variance
- Batch Failure
- Production Shortfall

**Expiry**

- Expiring Soon
- Expiring Today
- Expired

**Consumption**

- Abnormal Consumption
- Recipe Variance
- Unexpected Usage

**Waste**

- High Waste
- Repeated Waste
- Abnormal Waste Cost

**Replenishment**

- Requisition Pending
- Transfer Delayed
- Receiving Discrepancy
- Forecast Shortfall

**Availability**

- Product 86'd
- Modifier Unavailable
- Ingredient Unavailable

**Food Safety**

- Temperature Breach
- Quality Hold
- Failed Check

**Severity**

**Information | Warning | High | Critical**

**AI — Smart Replenishment**

TICVAI should not rely only on:

**Stock < Minimum → Reorder**

The recommendation engine should consider:

- Current Stock
- Available Stock
- Reserved Stock
- Incoming Stock
- Production Requirements
- Historical Consumption
- Forecast Demand
- Ticket Sales
- Reservations
- Events
- Safety Stock
- Supplier / Internal Lead Time

## Example

### Beef Patty

Available:  
980

Tomorrow Forecast:  
1,320

Safety Stock:  
80

Incoming:  
100

Projected Shortfall:

**320**

AI Recommendation:

Transfer 200 units from Central Kitchen and request an additional 120 units from Warehouse before 10:00 tomorrow.

Actions:

**Simulate | Create Requisition | Request Transfer | Ignore**

## AI — Production Shortfall Detection

Example:

Tomorrow Production Requirement:

Special Sauce:  
42 L

Available:  
12 L

Planned Production:  
30 L

Current production yield trend:

94%

Expected Actual Production:

28.2 L

Projected Available:

40.2 L

Shortfall:

**1.8 L**

TICVAI should identify this **before production is completed**.

Recommendation:

Increase today's production batch by approximately 3 L.

## **AI — Stock Redistribution**

Before recommending new procurement, TICVAI should look across internal stock locations.

Example:

Beach Kiosk:

Water:

Critical

Central Warehouse:

High Stock

VIP Lounge:

Overstock

TICVAI recommendation:

Transfer 120 units from VIP Lounge to Beach Kiosk rather than requesting additional stock.

This can reduce emergency purchasing and unnecessary stock holding.

## **AI — Waste & Consumption Anomaly Detection**

Example:

### **Chicken Breast**

Theoretical Consumption:

184 kg

Actual Reduction:

201 kg

Variance:

**+17 kg / 9.2%**

Estimated Value:

**AED 1,445**

AI identifies:

Consumption variance is significantly above the normal range.

Potential operational causes:

- Unrecorded Waste
- Portion Variance
- Recipe Configuration
- Production Yield
- Stock Count Error

Recommended:

### **Review Grill Station + Run Spot Count**

The AI should support investigation and should not automatically attribute discrepancies to employee misconduct.

## **AI — Expiry Optimization**

TICVAI should use expiry information to recommend stock movement.

Example:

### **Milk**

Main Restaurant:

40 L expires tomorrow

Beach Kiosk:

Projected requirement tomorrow:  
28 L

TICVAI recommendation:

Transfer 28 L from Main Restaurant to Beach Kiosk and prioritize this batch for consumption.

The inventory system should support:

### **FEFO — First Expired, First Out**

for expiry-sensitive items.

## AI — Production vs Demand

Board 5 should consume the production forecast created through Boards 2 and 6 and continuously compare:

### Forecast Demand

VS

### Production Plan

VS

### Actual Production

VS

### Actual Consumption

Example:

Burger Patties:

Forecast:  
1,480

Planned:  
1,550

Produced:  
1,510

Consumed:  
1,462

Remaining:  
48

This provides the feedback loop required to improve future forecasts.

## EMPLOYEE APP INTEGRATION

Board 5 shall integrate directly with the TICVAI Employee App.

Authorized employees should be able to perform:

- Stock Count
- Waste Entry
- Production Confirmation
- Batch Recording
- Transfer Dispatch
- Transfer Receipt
- Requisition
- 86 Product
- Restore Product

- Temperature Check
- Quality Check
- Receiving Confirmation

### Example Alert

#### Critical Stock — Beef Patty

Current:  
32 units

Expected depletion:  
1.4 hours

Nearby stock:

Central Kitchen:  
240

Action:

#### Request Transfer

### BARCODE / RFID SUPPORT

Where supported by the Shared Inventory Engine, Board 5 should allow employees to use:

- Barcode
- QR
- RFID

for:

Receiving

Stock Count

Transfer

Batch Identification

Waste Recording

Production Issue

Inventory Lookup.

The physical scanning capability should use the same inventory identifiers as the Shared Inventory module.

### BATCH, LOT & EXPIRY TRACEABILITY

Every relevant F&B ingredient and prepared product should support traceability.

Example:

**Beef Patty**

Batch:  
BP-140826-04

Supplier Batch:  
SUP-48292

Received:  
14 Aug

Expiry:  
18 Aug

Used in:

Main Restaurant

Production Batch:  
BURGER-PREP-0814

If a batch is placed on hold or recalled, TICVAI should be able to identify:

**Where it is****Which outlet received it****Which production batch used it**

and, where transaction data permits,

**which orders/products were affected.**

**APPROVAL & AUDIT CONTROLS**

Sensitive actions shall use TICVAI's shared Approval Engine.

Examples:

**Large Waste**

→ Manager Approval

**Inventory Adjustment**

→ Inventory Supervisor

**Emergency Transfer**

→ Authorized Role

**Production Variance**

→ Kitchen Supervisor

**Quality Hold Release**

→ Authorized Food Safety / Management Role

**Manual 86 Override**

→ Authorized F&B Role

All actions shall record:

**User**

**Timestamp**

**Location**

**Device**

**Action**

**Old Value**

**New Value**

**Reason**

**Approval**

**SHARED INVENTORY & PROCUREMENT BOUNDARY**

This architectural separation is mandatory.

**F&B BOARD 5 SHALL OWN**

- F&B Operational Stock Visibility
- Recipe Consumption
- Theoretical Consumption
- Production Execution
- Production Yield
- F&B Batch Usage
- Waste / Spoilage
- Outlet Stock Counts
- Operational Requisitions
- F&B Transfer Requests
- Outlet Receiving
- Product 86 / Availability
- Food-Safety Operational Checks
- F&B Stock Alerts
- AI F&B Replenishment Recommendations

**SHARED INVENTORY SHALL OWN**

- Inventory Item Master
- Enterprise Stock Ledger
- Warehouse Master

- Storage Locations / Bins
- Enterprise Batch / Lot Records
- Inventory Valuation
- Enterprise Goods Receipt
- Enterprise Transfer Transactions
- Inventory Adjustments
- Inventory Reconciliation
- Stock Valuation Methods

## PROCUREMENT SHALL OWN

- Supplier Master
- Supplier Contracts
- Purchase Requisition
- RFQ
- Supplier Quotations
- Quotation Comparison
- Purchase Order
- Supplier Delivery
- Purchasing Approval
- Supplier Invoice Matching
- Supplier Performance

Therefore, if an F&B outlet needs stock:

### F&B identifies requirement



### Board 5 creates operational replenishment request



Check Internal Availability



If available:

### Shared Inventory Transfer



If not available:

### Procurement Process



Goods Received



Shared Inventory Updated



F&B Availability Updated

This prevents duplicate inventory and purchasing systems.

**BOARD 5 — FINAL 10 SCREENS**

| #  | Screen                                               |
|----|------------------------------------------------------|
| 1  | F&B Stock & Operations Command Center                |
| 2  | Outlet Stock & Ingredient Availability               |
| 3  | Recipe Consumption & Theoretical Inventory           |
| 4  | Production Execution & Batch Management              |
| 5  | Wastage, Spoilage, Returns & Write-Off               |
| 6  | Requisition & Smart Replenishment                    |
| 7  | Transfers, Distribution & Outlet Receiving           |
| 8  | Stock Count, Reconciliation & Variance               |
| 9  | Product Availability, 86 & Operational Food Safety   |
| 10 | Operational Alerts, AI Replenishment & Action Center |

**COMPLETE BOARD 5 FLOW**

**F&B Sales + Reservations + Attendance Forecast**



**Board 2 Recipe & Production Plan**



Board 5 calculates operational requirements



**Production Execution**



Raw Ingredients Consumed



Prepared Stock Created



Distributed to Outlets



**Sales**



Recipe Explosion



Theoretical Consumption



Shared Inventory Updated



**Actual Stock Compared**



Variance / Waste / Expiry / Shortage Detection



TICVAI determines:

**Healthy Stock**

or

**Replenishment Required**



Check other internal locations



**Transfer Recommendation**

or

**Requisition**



Shared Inventory / Procurement



Stock Received



Outlet Availability Updated



Previously 86'd products identified



## Restore Product Availability



POS + Kiosk + QR + Mobile + B2C synchronized



Board 6 Analytics & AI receives operational results.

## ARCHITECTURAL REQUIREMENT

Board 5 shall **not maintain a separate F&B inventory ledger**.

There shall be one TICVAI inventory source of truth:

## F&B Operational Event



## Recipe / Production Consumption Engine



## Shared Inventory Ledger



## Availability & Forecasting



## Replenishment



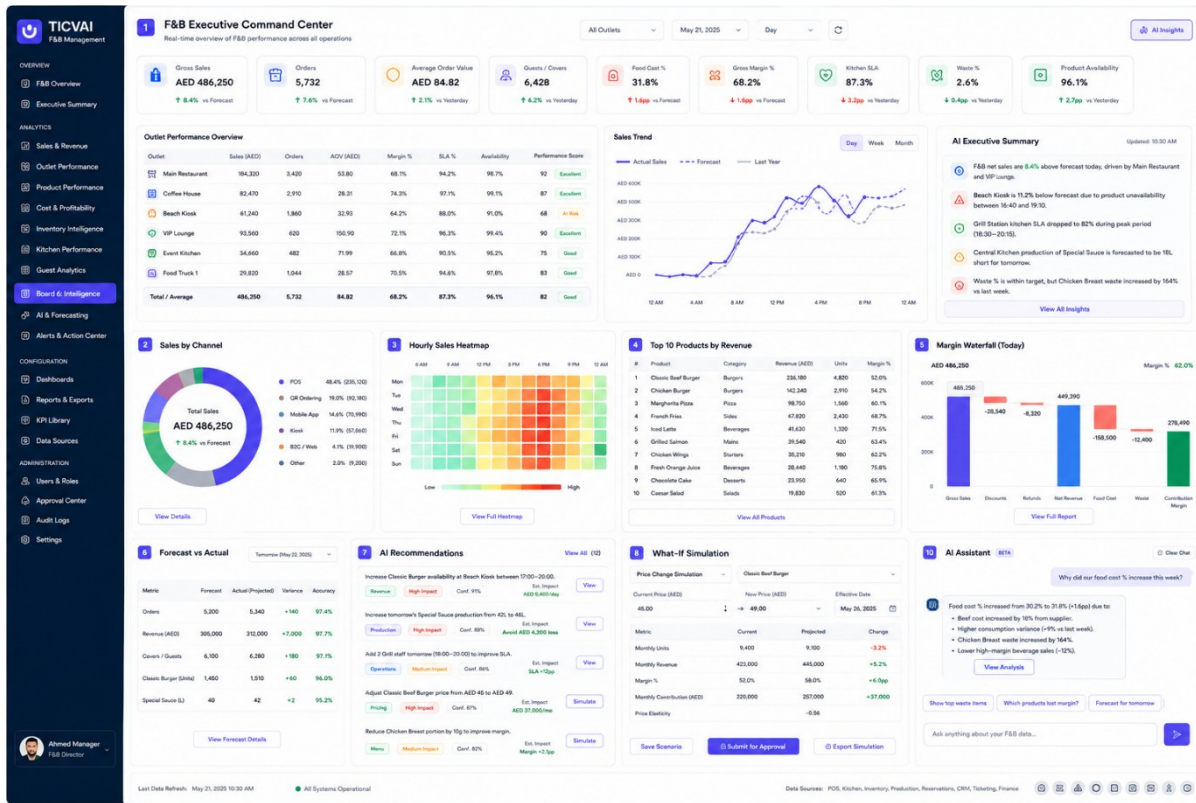
## Warehouse / Procurement

This same Shared Inventory Engine should later support **F&B, Retail, Rental, operational supplies and other applicable TICVAI modules**.

That gives TICVAI a clean architecture while Board 5 remains specialized enough to handle the operational realities of F&B: **recipes, production, batches, yield, waste, expiry, 86 management and food safety**.

# TICVAI F&B — BOARD 6

## F&B Intelligence, Analytics, Forecasting & AI Optimization — Scope of Work



### 1. Purpose

The F&B Intelligence, Analytics, Forecasting & AI Optimization Board shall provide a centralized management and decision-intelligence layer across all Food & Beverage operations.

The Board shall consolidate operational, commercial and financial information from:

- POS and Mobile POS
- QR Ordering
- Self-Service Kiosks
- B2C / Mobile Ordering
- Menus and Products
- Recipes and Ingredients
- Kitchen Display System (KDS)
- Kitchen Stations
- Tables and Reservations
- Guest Profiles / CRM
- Membership and Loyalty
- F&B Stock and Consumption
- Production and Central Kitchen
- Waste and Write-Offs
- Shared Inventory
- Procurement
- Resource Management

- Ticketing and Attendance
- Events and Experiences
- Payments
- Finance

The objective shall be to move beyond traditional reporting and provide a continuous intelligence cycle:

**Measure → Compare → Detect → Explain → Forecast → Recommend → Simulate → Approve → Act → Measure Results → Learn**

TICVAI shall help management answer:

**What happened?**

**Why did it happen?**

**What is likely to happen next?**

**What should we do about it?**

## SCREEN 1 — F&B Executive Command Center

---

### Purpose

Provide executives and F&B management with a real-time consolidated view of commercial performance, profitability, operations, inventory, production, kitchen performance and emerging risks.

### Global Filters

- Tenant
- Company
- Park
- Venue
- Outlet
- Outlet Type
- Date / Period
- Service Period
- Channel

### Primary KPI Cards

- Gross Sales
- Net Sales
- Orders
- Covers / Guests
- Average Order Value
- Revenue per Guest
- Food Cost %
- Gross Margin %
- Waste %
- Kitchen SLA %
- Average Preparation Time
- Product Availability %
- Stock-Out Events
- Forecast Accuracy

- Guest Satisfaction

### Comparisons

Every applicable KPI shall support:

- Today vs Yesterday
- Week vs Previous Week
- Month vs Previous Month
- Year vs Previous Year
- Actual vs Budget
- Actual vs Forecast
- Outlet vs Outlet
- Venue vs Venue

### Operational Health

Example:

**Main Restaurant** — Healthy

**Coffee House** — Healthy

**Beach Kiosk** — Stock Risk

**VIP Lounge** — Margin Warning

**Central Kitchen** — Production Risk

### AI Executive Summary

TICVAI shall automatically summarize significant developments.

Example:

**F&B net sales are 8.4% above forecast today. Main Restaurant is the primary contributor, while Beach Kiosk is 11.2% below forecast due primarily to product unavailability between 16:40 and 19:10. Grill SLA has declined to 82%, and Central Kitchen has a projected sauce-production shortfall for tomorrow.**

### Quick Actions

[View Revenue](#) | [View Risks](#) | [Forecast Tomorrow](#) | [AI Recommendations](#) | [Compare Outlets](#) | [Ask TICVAI](#)

## SCREEN 2 — Sales, Revenue & Channel Analytics

---

### Purpose

Provide detailed commercial analysis across all F&B sales channels.

### KPI Cards

- Gross Sales
- Net Sales
- Tax
- Service Charges
- Discounts
- Promotions
- Refunds
- Complimentary Value
- Orders
- Items Sold
- Average Order Value
- Items per Order

Revenue Analysis

Analyze by:

- Outlet
- Venue
- Product
- Product Category
- Menu
- Service Period
- Hour
- Day
- Channel
- Payment Method
- Guest Segment
- Membership Tier
- Promotion
- Event

Channel Performance

Compare:

- Fixed POS
- Mobile POS
- QR Ordering
- Kiosk
- Mobile App
- B2C
- Table Ordering
- Other Integrated Channels

Example:

| Channel | Revenue  | Orders | AOV  | Growth | Margin |
|---------|----------|--------|------|--------|--------|
| POS     | AED 235K | 4,820  | 48.8 | +6%    | 67%    |
| QR      | AED 92K  | 1,540  | 59.7 | +18%   | 70%    |
| Mobile  | AED 71K  | 1,130  | 62.8 | +24%   | 72%    |
| Kiosk   | AED 58K  | 1,410  | 41.1 | +9%    | 65%    |

Drill-Down

Users shall be able to navigate:

Venue

- Outlet
- Category
- Product
- Channel
- Transaction

without leaving the analytical workflow.

## SCREEN 3 — Outlet, Service & Guest Performance

### Purpose

Benchmark outlets and service models while connecting commercial performance with guest behavior.

### Outlet Scorecard

| Outlet          | Sales    | Orders | AOV   | Margin | SLA | Availability | Waste |
|-----------------|----------|--------|-------|--------|-----|--------------|-------|
| Main Restaurant | AED 184K | 3,420  | 53.8  | 68%    | 94% | 98%          | 2.1%  |
| Coffee House    | AED 82K  | 2,910  | 28.2  | 74%    | 97% | 99%          | 1.4%  |
| Beach Kiosk     | AED 61K  | 1,860  | 32.8  | 64%    | 88% | 91%          | 4.7%  |
| VIP Lounge      | AED 93K  | 620    | 150.0 | 72%    | 96% | 99%          | 2.8%  |

### Outlet Performance Score

TICVAI may calculate a configurable performance score based on:

- Revenue Performance
- Growth
- Margin
- Product Availability
- Kitchen SLA
- Waste
- Guest Satisfaction
- Service Performance

Example:

### Main Restaurant — 92 / 100

### Guest Analytics

Analyze:

- Unique F&B Guests
- Returning Guests
- Spend per Guest
- Orders per Guest
- F&B Conversion Rate
- Membership Spend
- VIP Spend
- Family Spend
- Group Spend

Guest information shall be consumed from TICVAI's shared CRM / Guest Profile rather than maintained separately within F&B.

### Ticket-to-F&B Conversion

A particularly important TICVAI metric:

### Park Attendance

12,500



F&B Customers

4,625

**F&B Conversion = 37%**

Management should be able to compare conversion by:

Venue

Ticket Type

Membership

Event

Time

Guest Segment.

---

## SCREEN 4 — Product Performance & Menu Engineering

---

**Purpose**

Analyze products using sales, popularity, profitability, recipe cost, waste, preparation performance and availability.

**Product KPIs**

- Units Sold
- Revenue
- Product Cost
- Gross Profit
- Margin %
- Popularity
- Average Preparation Time
- SLA %
- Waste Contribution
- Refund Rate
- Modifier Revenue
- Availability %
- Stock-Out Frequency

**Menu Engineering Matrix**

Products shall be categorized using configurable profitability and popularity thresholds:

**STAR**

High Popularity + High Profitability

**PLOWHORSE**

High Popularity + Low Profitability

**PUZZLE**

Low Popularity + High Profitability

**DOG**

Low Popularity + Low Profitability

**Product Example****Classic Beef Burger**

Units:

4,820

Revenue:

AED 236,180

Margin:

52%

Popularity:

High

Classification:

**PLOWHORSE**

### AI Insight

Classic Beef Burger remains one of the highest-volume products, but margin declined from 64% to 52%, primarily due to increased beef cost and higher-than-normal consumption variance.

### Recommended Actions

- Review Price
- Review Recipe
- Review Portion
- Review Supplier Cost
- Review Modifier Pricing
- Review Waste
- Simulate Price Change

## SCREEN 5 — Food Cost, Margin & Profitability Intelligence

---

### Purpose

Provide management with detailed understanding of F&B profitability from product level through enterprise level.

### KPI Cards

- Net F&B Revenue
- Theoretical Food Cost
- Actual Food Cost
- Food Cost %
- Gross Profit
- Gross Margin %
- Waste Cost
- Consumption Variance
- Discount Impact
- Complimentary Impact

### Theoretical vs Actual Cost

Example:

Net Revenue:

**AED 480,000**

Theoretical Food Cost:

**AED 142,000**

Actual Food Cost:

**AED 158,500**

Variance:

**AED 16,500**

### Profitability Dimensions

Analyze by:

- Tenant
- Venue
- Outlet
- Product
- Category
- Recipe
- Ingredient
- Kitchen
- Service Period
- Channel
- Promotion

### Margin Waterfall

#### Gross Sales



Discounts



Refunds



Net Revenue



Theoretical Food Cost



Consumption Variance



Waste



#### Contribution Margin

#### Cost Change Impact

When ingredient cost changes, TICVAI should identify affected products.

Example:

#### Beef Cost +18%

Affected Products:

12

Average Margin Impact:

-5.8%

Potential Monthly Impact:

**AED 84,000**

This information should use cost and finance information from the shared Inventory, Procurement and Finance layers.

---

## SCREEN 6 — Inventory, Waste & Production Intelligence

### Purpose

Transform Board 5's operational stock, production, consumption and waste information into management intelligence.

### KPI Cards

- Inventory Consumption
- Theoretical Consumption
- Actual Consumption
- Consumption Variance
- Waste Value
- Waste %
- Expired Stock
- Production Yield
- Production Variance
- Stock-Out Events
- 86 Events
- Estimated Lost Sales

**Production Analytics**

Compare:

**Forecast Demand**

vs

**Production Plan**

vs

**Actual Production**

vs

**Actual Consumption**

Example:

| Item          | Forecast | Planned | Produced | Consumed | Variance |
|---------------|----------|---------|----------|----------|----------|
| Burger Patty  | 1,480    | 1,550   | 1,510    | 1,462    | +48      |
| Pizza Dough   | 680      | 720     | 704      | 676      | +28      |
| Special Sauce | 38 L     | 42 L    | 39 L     | 37 L     | +2 L     |

**Production Yield Analysis**

Example:

Special Sauce

Expected Yield:

100 L

Actual:

94 L

Yield:

94%

Historical:

97%

TICVAI should highlight abnormal yield.

**Waste Analysis**

Analyze by:

- Ingredient
- Product

- Recipe
- Outlet
- Kitchen
- Station
- Reason
- Shift
- Service Period
- Production Batch

### Lost Sales Analysis

If a product is unavailable:

Classic Burger unavailable:

16:40–18:55

Expected Units:

148

Actual Units:

24

Estimated Lost Sales:

**AED 6,076**

Estimated Lost Margin:

**AED 3,460**

This allows management to understand the commercial impact of stock and production problems.

## SCREEN 7 — Kitchen, Service & Fulfilment Performance

---

### Purpose

Combine Board 3 kitchen information and Board 4 restaurant-service information into one management-performance view.

### Kitchen KPIs

- Orders Prepared
- Average Preparation Time
- Kitchen SLA %
- Delayed Orders
- Re-Fire %
- Station Utilization
- Peak Kitchen Load
- Fire-to-Ready Time
- Ready-to-Collection Time

### Service KPIs

- Reservations
- Walk-Ins
- Covers
- Average Wait Time
- Seating Time
- Dining Duration

- Table Turnover
- Revenue per Table
- Revenue per Cover
- No-Show %
- Guest Satisfaction

### **Kitchen Bottleneck Analysis**

Example:

#### **Grill Station**

Target:

10 min

Actual:

14.8 min

SLA:

76%

Peak Problem:

18:30–20:15

### **Course Firing Analytics**

Measure:

- Average Hold Time
- Fire-to-Ready
- Course Gap
- Late Fires
- Manual Overrides
- Timed-Fire Accuracy

### **Collection Analytics**

Measure:

- Ready-to-Collection Time
- Buzzer Response
- Digital Notification Response
- Orders Waiting After Ready

### **Guest Service Journey**

TICVAI should measure:

#### **Guest Arrival**

→ Wait

→ Seating

→ Order

→ Fire

→ Preparation

→ Service

→ Dining

→ Bill Request

→ Payment

This allows management to identify exactly where service delays occur.

## SCREEN 8 — Demand Forecasting & Operational Planning

---

### Purpose

Forecast future F&B demand and convert it into actionable plans for products, ingredients, production, stock, kitchens and staffing.

### Forecast Inputs

The forecasting engine should be capable of using:

- Advance Ticket Sales
- Attendance Forecast
- Event Schedule
- Restaurant Reservations
- Historical F&B Sales
- Historical Conversion
- Membership Visits
- Seasonality
- Day of Week
- Service Period
- Promotions
- Holidays
- School / Group Bookings
- Weather, where integrated
- Current Stock
- Incoming Stock

### Example Forecast

#### Tomorrow — Dubai Park

Expected Attendance:

**14,800**

Expected F&B Conversion:

**36%**

Expected Orders:

**5,328**

Expected Revenue:

**AED 312,000**

#### Product Forecast

Classic Burger:

1,480

Chicken Burger:

920

Pizza:

680

Coffee:

2,150

Water:

3,840

**Ingredient Requirement**

Beef Patties:

1,550

Burger Buns:

2,620

Milk:

210 L

Fries:

420 kg

**Production Requirement**

Special Sauce:

42 L

Pizza Dough:

720

Marinated Chicken:

280 kg

**Kitchen Capacity Forecast**

Grill:

108%

Fryer:

94%

Beverage:

81%

**Staffing Requirement**

Recommended:

+2 Grill Staff

+1 Expeditor

+2 Cashiers

+1 Runner

Staffing recommendations shall integrate with TICVAI Resource Management.

---

## SCREEN 9 — AI Recommendation, Simulation & Optimization Center

---

**Purpose**

Provide a centralized AI-driven recommendation engine that identifies commercial and operational opportunities and allows management to simulate outcomes before taking action.

**Recommendation Categories**

- Revenue
- Pricing
- Margin
- Menu
- Product Availability
- Production
- Inventory
- Waste
- Kitchen

- Staffing
- Guest Experience

**Recommendation Card**

Each recommendation should display:

**Issue / Opportunity****Business Impact****Root Cause****Recommended Action****Confidence****Expected Result****Affected Locations****Example — Revenue Opportunity**

Increase Classic Burger availability at Beach Kiosk between 17:00–20:00.

Estimated additional daily revenue:

**AED 8,400**

Confidence:

**91%**

Reason:

High demand combined with recurring stock-outs.

**Example — Production**

Increase tomorrow's Special Sauce production from 42 L to 46 L.

Reason:

Current production yield is averaging 94%, creating a forecast shortfall.

**Example — Staffing**

Grill Station is forecast to exceed 100% capacity between 18:00–20:30.

Recommendation:

**Add 2 Grill Staff and activate Grill B at 17:45.**

**What-If Simulation**

Authorized users shall be able to simulate changes before approval.

Example:

What happens if Beef Burger price changes from AED 45 to AED 49?

**Simulation**

Current Monthly Units:

9,400

Estimated Demand Change:

-3.2%

Current Revenue:

AED 423,000

Projected Revenue:

AED 445,000

Current Margin:

52%

Projected Margin:

58%

Estimated Contribution Improvement:

**+AED 37,000/month**

Actions:

**Save Scenario | Modify | Submit for Approval**

### **Production Simulation**

Example:

What if attendance tomorrow reaches 18,000 instead of 14,800?

TICVAI recalculates:

Orders

Product Demand

Ingredient Requirements

Production Requirements

Kitchen Capacity

Staffing

Replenishment.

### **Recommendation Controls**

Actions:

**View Analysis**

**Simulate**

**Accept**

**Modify**

**Reject**

**Assign**

**Submit for Approval**

AI shall remain advisory for material commercial, financial or operational decisions unless an authorized tenant explicitly configures an automated rule.

## **SCREEN 10 — AI Assistant, Alerts & Management Action Center**

---

### **Purpose**

Provide a conversational management interface for querying F&B data, investigating issues and initiating controlled business actions.

### **Natural-Language Questions**

Authorized users should be able to ask:

How did F&B perform yesterday?

Why did our food cost increase this week?

Which outlet has the highest waste?

Which products are causing kitchen delays?

Which products are likely to run out tomorrow?

Why is Beach Kiosk underperforming?

How much revenue did we lose because of 86'd products?

Which production batches had poor yield?

Which outlet has the lowest ticket-to-F&B conversion?

What should Central Kitchen produce tomorrow?

How many kitchen staff do we need this weekend?

### Example Response

**Yesterday F&B generated AED 486,250, 8.4% above forecast.**

Main Restaurant contributed most of the increase. Beach Kiosk underperformed by 11.3%, primarily because five products were unavailable during peak service.

Estimated revenue lost due to unavailable products was AED 7,850.

Grill Station also missed SLA on 18% of orders between 18:30 and 20:00.

### Suggested Actions

**View Beach Kiosk**

**Review Stock-Outs**

**Create Replenishment**

**Review Grill Performance**

### NATURAL-LANGUAGE CONTROLLED ACTIONS

The AI Assistant should also allow authorized users to propose actions conversationally.

Example:

Increase Classic Burger price to AED 49 next Monday.

TICVAI shall not immediately execute the request.

It should create:

### Proposed Change

Product:

Classic Burger

Current Price:

AED 45

Proposed:

AED 49

Effective:

Next Monday

Affected:

6 Outlets

Channels:

POS + QR + Mobile + Kiosk

Current Margin:

52%

Projected:

58%

Then:

**Simulate | Edit | Submit for Approval | Cancel**

The action shall follow TICVAI's standard approval workflow.

### AI ROOT-CAUSE ANALYSIS

TICVAI should correlate data across modules rather than only report symptoms.

Example:

**Issue**

Beach Kiosk Revenue:

**-11.3% vs Forecast****AI Analysis**

Primary Cause:

Product Availability

Five high-volume products unavailable for 2h 30m.

Underlying Cause:

Burger Bun shortage.

Underlying Cause:

Requisition submitted 3.2 hours late.

Secondary Cause:

Grill SLA deterioration during peak period.

**Estimated Impact**

Lost Revenue:

AED 7,850

Lost Margin:

AED 4,210

**Recommended Actions**

1. Increase Burger Bun safety stock.
2. Enable forecast-based replenishment.
3. Move requisition trigger 4 hours earlier.
4. Review Grill peak staffing.

This cross-module analysis should be a major TICVAI capability.

**AI EXPLAINABILITY**

Every significant AI recommendation shall provide an explanation.

Example:

**Recommendation**

Increase Classic Burger:

AED 45 → AED 49

**Why?**

Ingredient Cost:

+18%

Current Margin:

52%

Target Margin:

60%

Demand Trend:

Strong

Price Sensitivity Estimate:

Low-Moderate

**Confidence**

87%

Data Used

- Product Sales History
- Recipe Cost
- Ingredient Cost
- Promotion History
- Historical Price Changes
- Current Demand

Managers must be able to understand the basis of significant recommendations.

FORECAST ACCURACY & MODEL PERFORMANCE

TICVAI shall measure the performance of its own forecasts.

| Forecast          | Predicted | Actual   | Accuracy |
|-------------------|-----------|----------|----------|
| Orders            | 5,200     | 5,340    | 97.4%    |
| Burger Sales      | 1,450     | 1,510    | 96.0%    |
| Revenue           | AED 305K  | AED 312K | 97.7%    |
| Sauce Requirement | 40 L      | 42 L     | 95.2%    |

Users should be able to analyze forecast accuracy by:

- Outlet
- Product
- Day
- Forecast Horizon
- Service Period
- Event

The system should retain forecast versions so management can compare **what was predicted at the time** against actual results.

RECOMMENDATION OUTCOME TRACKING

TICVAI shall measure whether accepted recommendations produced the expected outcome.

Example:

Recommendation

Add 2 Grill Staff from 18:00–20:00.

Before

Average Prep:

14.2 min

Kitchen SLA:

78%

After

Average Prep:

10.8 min

Kitchen SLA:

93%

Result

Preparation Time:

**Improved 24%**

The intelligence loop becomes:

**Detect → Recommend → Approve → Execute → Measure → Learn**

**ANOMALY DETECTION**

TICVAI should proactively identify unusual operational or commercial patterns.

Examples:

- Unusual Sales Drop
- Unusual Refund Increase
- Abnormal Discounting
- Abnormal Complimentary Items
- Consumption Variance
- Waste Spike
- Production Yield Drop
- Kitchen SLA Deterioration
- Stock-Out Pattern
- Unusual Product Demand
- Unexpected Margin Change

Example:

**Waste Alert — Chicken Breast**

Current:

8.2%

Historical:

3.1%

Deviation:

+164%

Most affected:

Main Restaurant — Grill Station — Dinner Shift

The system should support investigation without automatically attributing anomalies to employee misconduct.

**PRODUCTION & CENTRAL KITCHEN INTELLIGENCE**

Board 6 shall provide specific analytics for central kitchens and commissaries.

**KPIs**

- Production Plan Completion %
- Planned vs Actual Production
- Production Yield
- Batch Failure Rate
- Cost per Production Unit
- Production Waste
- Outlet Demand Fulfilment
- On-Time Production
- Transfer Fulfilment

**AI Example**

Central Kitchen is forecast to miss tomorrow's pizza dough requirement by approximately 8%.

Reason:

Current yield below target + higher-than-forecast event demand.

Recommendation:

**Increase production run by 60 units.**

#### **CROSS-SELL & UPSELL INTELLIGENCE**

TICVAI should analyze product combinations to identify commercial opportunities.

Example:

Guests purchasing:

##### **Classic Burger**

also purchase:

Fries — 78%

Soft Drink — 72%

Dessert — 18%

AI recommendation:

Create a Burger + Fries + Drink combo at AED 62.

Estimated attach-rate improvement:

+14%

Estimated monthly revenue impact:

**+AED 42,000**

The recommendation should proceed through simulation and approval before menu publication.

#### **TICKETING + F&B INTELLIGENCE**

This should be a major differentiator for TICVAI.

The F&B intelligence layer should correlate food demand with attraction activity.

Example:

##### **Waterpark**

Expected Attendance:

8,400

Historical F&B Conversion:

41%

Expected F&B Guests:

3,444

Expected Peak:

13:00–15:30

TICVAI recommends:

- Increase Beach Kiosk stock
- Prepare additional cold beverages
- Increase lunch production
- Add 2 cashiers
- Activate second fryer

A standalone restaurant system generally does not have the same direct access to ticketing, attendance, experience and attraction demand signals.

#### **ROLE-BASED INTELLIGENCE**

Board 6 shall use TICVAI's common RBAC/PBAC framework.

#### **Executive / CEO**

- Revenue
- Margin
- Growth
- Forecast
- Business Risks
- AI Opportunities

**F&B Director**

- Outlet Performance
- Menu
- Margin
- Waste
- Production
- Kitchen
- Forecast

**Restaurant Manager**

- Sales
- Kitchen
- Tables
- Guest Service
- Stock
- Staff

**Executive Chef**

- Kitchen
- Recipes
- Production
- Yield
- Food Cost
- Waste
- Ingredient Availability

**Finance**

- Revenue
- Margin
- Discounts
- Refunds
- Cost
- Variance

**Inventory Manager**

- Consumption
- Stock
- Production Requirements
- Waste
- Forecast
- Replenishment

Users shall only see data and actions permitted by their role and context.

### **CONFIGURABLE DASHBOARD BUILDER**

Authorized users shall be able to configure dashboards without software development.

#### **Widget Library**

Examples:

- Revenue
- Orders
- Average Order Value
- Food Cost
- Margin
- Waste
- Kitchen SLA
- Stock
- 86 Events
- Table Occupancy
- Production
- Forecast
- AI Insight
- Alerts

Users shall be able to:

#### **Add Widget**

#### **Drag**

#### **Resize**

#### **Filter**

#### **Save**

#### **Share**

Dashboards may be:

- Private
- Role-Based
- Outlet-Based
- Venue-Based
- Shared
- Executive

### **REPORTING, EXPORT & SCHEDULING**

Reports and analytical views shall support appropriate:

- Excel Export
- CSV Export
- PDF Export
- Scheduled Reports
- Email Distribution
- Saved Filters
- Saved Views

Example:

### **Weekly F&B Executive Report**

Schedule:

Every Monday — 08:00

Recipients:

CEO

F&B Director

Finance Director

Contents:

Revenue

Margin

Food Cost

Waste

Kitchen SLA

Production

Forecast

Top Risks

AI Recommendations.

All reports shall respect tenant, venue and role permissions.

### **APPROVAL CENTER INTEGRATION**

AI and analytical recommendations that result in controlled changes shall integrate with TICVAI's common Approval Engine.

Examples:

#### **Product Price Change**

→ Commercial / Finance Approval

#### **Production Increase**

→ F&B / Kitchen Approval

#### **Emergency Stock Transfer**

→ Inventory Approval

#### **Large Waste Adjustment**

→ Management Approval

#### **Staffing Recommendation**

→ Resource / Operations Approval

#### **Menu Change**

→ F&B Approval

Actions:

**Approve | Reject | Request Changes | Delegate**

Board 6 shall not create a separate F&B approval engine.

### **AUDITABILITY**

The platform shall maintain audit records for significant analytical and AI-driven actions.

Records should include:

- Recommendation ID
- Recommendation Date
- AI / Rule Version where applicable
- Data Context

- User
- Decision
- Simulation
- Approval
- Executed Action
- Previous Value
- New Value
- Result
- Timestamp

This allows management to understand not only **what changed**, but also **why the change was proposed and who approved it**.

**BOARD 6 — FINAL 10 SCREENS**

| #  | Screen                                              |
|----|-----------------------------------------------------|
| 1  | F&B Executive Command Center                        |
| 2  | Sales, Revenue & Channel Analytics                  |
| 3  | Outlet, Service & Guest Performance                 |
| 4  | Product Performance & Menu Engineering              |
| 5  | Food Cost, Margin & Profitability Intelligence      |
| 6  | Inventory, Waste & Production Intelligence          |
| 7  | Kitchen, Service & Fulfilment Performance           |
| 8  | Demand Forecasting & Operational Planning           |
| 9  | AI Recommendation, Simulation & Optimization Center |
| 10 | AI Assistant, Alerts & Management Action Center     |

**COMPLETE BOARD 6 INTELLIGENCE FLOW**

**Operational Data**

POS + QR + Mobile + Kiosk



Menus + Products + Recipes



Kitchen + KDS + Course Firing



Tables + Reservations + Guest Service



Production + Stock + Consumption + Waste



Ticketing + Attendance + Events



CRM + Loyalty + Membership



Resource Management + Finance



**TICVAI F&B INTELLIGENCE LAYER**

**Collect**



**Normalize**



**Analyze**



**Compare**



**Detect Anomalies**



**Explain Root Cause**



**Forecast**



**Recommend**



**Simulate**



**Approval**



**Execute through Source Module**



**Measure Outcome**



**Improve Future Recommendations**

#### **ARCHITECTURAL REQUIREMENT**

Board 6 shall be implemented using TICVAI's common **Analytics, Forecasting & AI Intelligence Layer**, rather than as an isolated collection of F&B reports.

The architecture shall follow:

#### **Operational Systems**

F&B + Retail + Ticketing + Inventory + Procurement + CRM + Resource Management + Finance



#### **Unified Data & Analytics Layer**



#### **KPI / Reporting Engine**



#### **Forecasting Engine**



#### **Anomaly & Root-Cause Engine**



#### **AI Recommendation Engine**



#### **Simulation / What-If Engine**



#### **Approval & Action Engine**



#### **Operational Systems**



## Outcome Measurement

This shared architecture shall later allow **Retail, Shared Inventory & Procurement and other TICVAI modules to use the same intelligence infrastructure**, while Board 6 provides the specialized F&B KPIs, forecasts, recommendations and management workflows required for Food & Beverage operations.

The F&B module is therefore completed through six connected boards:

**Board 1 — Outlet & Operational Configuration**

**Board 2 — Menu, Product, Recipe & Production Configuration**

**Board 3 — Kitchen, KDS & Order Fulfilment**

**Board 4 — Tables, Reservations & Service Management**

**Board 5 — Stock, Production, Replenishment & Operational Control**

**Board 6 — Intelligence, Analytics, Forecasting & AI Optimization.**

# ADDITIONAL — STANDARD F&B POS FUNCTIONAL COVERAGE

**Purpose.** This amendment adds the POS operational functions identified in the F&B POS Standard Functions document that were missing or not stated explicitly in the backend structure. These requirements extend the existing six-board architecture and do not replace the more advanced capabilities already defined in the document.

**Implementation principle.** The functions below shall use the existing shared services for identity and access, payments, finance, CRM, inventory, approvals, audit, analytics and device management. Configuration belongs mainly to Board 1; kitchen execution to Board 3; table and guest service to Board 4; and reporting and intelligence to Board 6.

## A. Functional Summary

| Function Group      | Functions Added / Clarified                                                             | Primary Placement    |
|---------------------|-----------------------------------------------------------------------------------------|----------------------|
| Access and shifts   | Opening float, cashier cash-in/cash-out, suspend/resume and explicit shift controls     | Board 1              |
| Check lifecycle     | Check numbering, recall/inquiry, open-check list, preview and customer-check printing   | Boards 1 and 4       |
| Order and KOT       | Incremental KOT printing, KOT history/reprint and detailed item-entry behaviour         | Boards 1 and 3       |
| Corrections         | Detailed item/check void and item-transfer transaction rules                            | Boards 1 and 4       |
| Table service       | Cover-wise ordering, service-status marking and configurable visual table states        | Boards 3 and 4       |
| Delivery guests     | Telephone-led delivery profiles, addresses, preferences and order history               | Board 4 / Shared CRM |
| Pricing and payment | Manual happy hour, detailed tenders, tips, discounts, account charges and tax exemption | Boards 1 and 4       |
| Reporting           | Cashier, master, daily operational and legacy MIS report catalogue                      | Board 6              |
| POS usability       | Touch/mouse operation, shortcut areas, keypad, prompts and check-header configuration   | Board 1              |

## B. Board 1 Additions — User Access, Shift and Till Controls

Board 1 shall add a Cashier, Shift and Till Configuration area with the following standard functions:

- Configure mandatory opening cash balance/imprest by outlet, till, cashier and shift.
- Support cashier cash-in and cash-out with date/time, outlet, terminal, shift, opening balance, closing declaration and approval status.

- Track all transactions by cashier and shift and retain the applicable terminal and outlet context.
- Allow an authenticated cashier to suspend a POS session temporarily and resume it using password or approved authentication.
- Configure automatic return to the cashiering screen and timeout/session-lock rules.
- Configure a default outlet for authorized users while allowing permitted users to display and switch between available outlets.
- Display the active outlet, cashier, terminal and shift in the POS header.
- Provide user-level and shift-level reporting and a complete audit trail of opening, suspension, resumption and cash-out events.

## C. Board 1 and Board 4 Additions — Check and Order Lifecycle

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The Unified F&B Order Engine shall explicitly support the complete standard check lifecycle:

- Create a check with an automatically generated unique check number.
- Capture table number, cover count and server/waiter; allow configured defaults for quick-service and fast-food operations.
- Support table-service, counter, takeaway and delivery checks using the same order engine.
- Save an open check and retain all unsaved, saved, fired, printed and settled states.
- Recall a check by check number, table number or server number and show a selection list where several checks match.
- List all open checks for the current cashier and outlet, with permission-based access to other cashiers' checks.
- Preview the customer check before printing and support check inquiry by outlet and check number.
- Recall printed checks from the settlement workflow without duplicating or losing the original transaction reference.
- Print customer checks to configured continuous-feed, cut-sheet or receipt printers and prompt for manual paper insertion where required.
- Add individual items or numeric quantities, display description and price, scroll through the order and recalculate totals immediately.
- Support menu lists, category/submenu navigation, menu-item shortcut keys and configurable POS layouts already defined in Board 2.

## D. Board 1 and Board 3 Additions — KOT Management

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The existing KDS and printer-routing architecture shall include the following Kitchen Order Ticket controls:

- Automatically print or transmit a KOT when a check is saved or fired, according to the configured routing policy.
- On subsequent saves, send only newly added, changed or previously unsent items unless an authorized full reprint is requested.
- Maintain KOT history against the check and allow users to view all KOTs associated with a check or table.
- Display multiple KOTs separately, including sequence, station, print/fire time, user, status and reprint indicator.
- Reprint a KOT by outlet and check number, subject to permission, reason capture and audit logging.
- Print or display course-fire and course-hold instructions and reflect kitchen preparation status in the live table layout.

## E. Board 1 and Board 4 Additions — Void and Item Transfer Controls

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Void and transfer actions shall extend the existing approval and audit framework with explicit transaction behaviour:

- Void selected items, require a configured void reason and retain the original item plus a linked reversal/negative entry.
- Display a clear void indicator, recalculate the check balance and save or print the correction as configured.
- Void an entire check by reversing every original item, reducing the balance to zero and retaining the original and reversal audit chain.
- Report item voids and check voids by outlet, cashier, server, terminal, reason, approver, date and value.
- Move selected items from a source check to an existing or newly created destination check.
- Locate the destination by table, check or server and display all matches where the reference is not unique.
- Update both source and destination totals atomically and retain user, reason, approval and cross-referenced check numbers.

## **F. Board 3 and Board 4 Additions — Cover, Course and Service Status**

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The existing course, kitchen and table-service functionality shall be expanded or clarified as follows:

- Assign ordered items to individual covers/guests and show the cover number against each item.
- Allow cover-wise and course-wise ordering within the same check.
- Create multiple courses, tag items by course, prompt for the first course and manually fire or hold later courses.
- Mark one or more items as serviced and display the service status against each item and in the table layout.
- Provide configurable colour-coded table states including vacant, opened/no order, preparing, partially ready, ready/serviced and cleaning.
- Provide a visible status legend and permit authorized configuration of table-state colours.
- Open the check and KOT history directly from the table map, including separate views for multiple KOTs.

## **G. Board 4 Additions — Delivery Outlet and Guest Profile Functions**

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Delivery outlets shall use the shared TICVAI Guest Profile/CRM while supporting the following POS workflow:

- Configure an outlet as delivery-enabled and permit telephone number to act as the primary order lookup instead of table number.
- Search existing guests by telephone number and prompt to create a profile when no match is found.
- Capture guest name, full address, multiple contact numbers, birthday, anniversary, guest class and preferences.
- Edit the guest profile subject to CRM permissions and data-protection controls.
- Automatically retrieve returning-guest details and display previous orders and the items within a selected order.
- Filter or group order history by outlet and other permitted dimensions.
- Use the standard order, KOT, settlement and reporting workflows for delivery checks.

## **H. Board 1 and Board 4 Additions — Happy Hour, Discounts, Tips, Tax and Settlement**

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The pricing and settlement design shall explicitly include these standard operational controls:

- Permit authorized manual start/end of happy-hour pricing, display a header indicator and apply configured happy-hour prices only to subsequently posted items.
- Configure whether manual happy-hour switching is allowed and report normal-hour and happy-hour sales separately.
- Support cash settlement with exact payment, over-tender and automatic change calculation.

- Display configured card types and capture only PCI-compliant payment references or masked/tokenized details required by the payment integration.
- Support room, guest, special, group, corporate and city-ledger account charges where integrated, including room/account inquiry and advanced search.
- Support partial and mixed-tender settlement, with the outstanding balance recalculated after each tender.
- Add, confirm and remove a manual tip before final settlement and include it in the check total and tip reporting.
- Apply predefined or ad hoc percentage discounts to the entire check, all revenue heads or selected heads/categories.
- Capture guest and membership references where required, remove an applied discount and recalculate the balance.
- Apply configured full or partial tax-exemption categories and recalculate the check balance while retaining approval and audit details.

## I. Board 4 Addition — Standard Split-Check Behaviour

In addition to split-by-guest and split-by-item, the system shall support the following standard split-check controls:

- Split one check into multiple independent checks, with a configurable limit that may be restricted to the number of covers.
- Divide the balance equally by default and allow authorized custom amounts.
- Validate that the total of all split checks exactly equals the original balance.
- Recall and settle each split check independently using different payment methods.
- Maintain split-check lineage and separate reporting while preventing unsupported changes to the original check after completion of the split.

## J. Board 6 Additions — Standard POS Report Catalogue

Board 6 shall include, or source through the common reporting layer, the following standard operational reports:

- Cashier reports: Shift End, Daily Summary, Item Sales, Open Checks, Check Inquiry, Reprint KOT and Reprint Check.
- Daily reports: Day Summary, Settlement, Item Sales, Open Checks, Sales by Check, Settlement by Check and Covers.
- Control reports: Void Items, Void Checks, Discounts, Shift End Summary, Tax-Exempted Checks, 100% Discounted Checks, Discounts by Category, Authorization, Cash Drop, Tax Scroll, Tips Scroll and Reprint Check Logs.
- MIS reports: Charge Summary, Covers Summary, Item Sales Summary, Top 10 Best/Worst Selling Items, Item Sales by Day of Week, Contact List, MIS Void Items, NC Check Detail and Loyalty Card Details.
- To-date MIS summaries: Item Sales, Charge, Covers, Settlement, Discount and Tax, with MTD and YTD values where applicable.
- Report filters: date/date range, time range, tenant, company, venue, outlet, cashier, server, shift, terminal, settlement method, item group, item, product type, menu, channel, normal/happy hour and inclusion/exclusion of open checks.
- Report output to screen, configured printer and approved export formats, subject to role-based permissions.
- Item Sales reporting shall include quantity, sales value and potential/theoretical cost.